

Experimental Evidence for the HFT Arms Race: Welfare Decomposition and Market Fragility on a Formally Verified Limit Order Book

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Abstract

We provide an experimental test of the Budish–Cramton–Shim (BCS) theory of the high-frequency trading (HFT) arms race, conducted on a continuous limit order book whose matching core has been model-checked in TLA+ with zero invariant violations. Driving latency-differentiated snipers, an adverse-selection market maker, and noise traders through the verified engine via a pybind11 bridge, we decompose realized welfare into market-maker, noise-trader, and HFT components and find that HFT latency-arbitrage rent is a transfer away from liquidity providers—borne almost entirely by the market maker—accompanied by significantly more liquidity gaps. The decomposition is closed rather than assumed: the conservation residual is indistinguishable from floating-point noise (mean 1.6×10^{-13}), an internal-consistency check confirming the harness leaks no cash or inventory rather than an economic result in itself. We then trace the mechanism across three further sweeps: rent scales approximately linearly in market-maker staleness with no detectable phase transition; as the number of HFTs grows, per-HFT rent dissipates as roughly $1/k$ while total extracted rent stays flat, so the marginal social cost of additional speed competition surfaces as rising fragility rather than rising rent; and frequent batch auctions recover market-maker welfare monotonically away from the degenerate single-tick interval (about 97% recovery by an interval of 500), with point estimates of rent also falling though more noisily. Because the matching layer is model-checked against its specification—at a bounded scope, with zero invariant violations, and now including the continuous matching/cross action itself—the emergent arms-race dynamics are attributable to agent incentives rather than to matching-engine error, isolating them as consequences of rational latency competition on the continuous-clearing design itself.

1. Introduction

1.1 The arms-race argument

Budish, Cramton, and Shim (2015, hereafter BCS) argue that the continuous limit order book—the dominant design of modern equity and futures markets—contains a structural flaw that has nothing to do with any individual participant’s conduct. Because the book processes messages serially in continuous time, each arrival of public information creates a fleeting, mechanical arbitrage opportunity: whichever trader reacts first can pick off the stale quotes of liquidity providers before those quotes can be revised. The competition to be first is a contest over speed rather than price, and in equilibrium it dissipates real resources into a socially wasteful arms race while transferring rent from liquidity providers (and, indirectly, the investors they serve) to the fastest “snipers.” Crucially, the resulting latency tax widens effective spreads and degrades liquidity without

improving price discovery; it is a prisoner’s dilemma baked into the market’s microstructure. BCS propose frequent batch auctions—discrete-time, uniform-price clearing—as a design remedy that converts competition on speed back into competition on price, neutralizing the mechanical sniping advantage.

1.2 Why a formally verified substrate matters

Testing this argument by simulation confronts a methodological problem that has dogged agent-based market models since their inception: when an emergent pathology appears, one cannot easily tell whether it reflects the economics under study or an artifact of the bespoke, unverified matching machinery on which the agents trade. A subtle bug in order-priority handling, partial-fill accounting, or event sequencing can manufacture exactly the kind of liquidity gaps and welfare transfers a researcher hopes to observe. Our contribution to internal validity is to run the entire BCS mechanism—latency-differentiated observation of a common fundamental value and a submission-timing race—as a Python scheduler sitting atop a C++ limit order book whose matching core has been model-checked in TLA+, accessed through a pybind11 bridge. Because the matcher is deterministic and model-checked against its specification (at bounded scope, with zero invariant violations, and now including the cross action that produces fills), any emergent arms-race dynamics, liquidity evaporation, or crash behavior we report are attributable to agent incentives rather than to design-level matching error; the precise reach of that guarantee is stated at the end of this subsection. This pairing of a verified substrate with a closed, auditable welfare-accounting identity is, to our knowledge, novel in the market-microstructure literature.

We report the verification provenance precisely. The engine’s matching specification (`MatchingEngine.tla`) is model-checked by TLC with zero invariant violations. As of this revision the specification models the continuous matching/cross action itself—price-cross execution and FIFO selection of the earliest-timestamped resting order—checking the `FIFOExecution` and `MatchingConservation` invariants alongside the book-management invariants (`NoNegativeQuantity`, `FIFO_Preservation`, `GTD_Expiry_Correctness`); at its bounded scope (`MaxOrders=3`, `MaxTime=2`, `Prices={100,200}`) this run explores on the order of 1.3×10^6 distinct reachable states with zero violations. (An earlier configuration of the same spec explored a far larger space—on the order of 4.5×10^8 generated states—but modelled only book management, without the cross action; the matching-inclusive model is necessarily smaller because adding execution forces tighter bounds to remain exhaustive.) These counts are engine-verification artifacts recorded with the spec, not experimental results of this paper, and we cite them as such rather than as findings of the experiments below. For the session-lifecycle specification (`Auction.tla`, §3.1) we did not recover a model-checked state count from the spec tree and decline to report one we cannot verify. The substantive claim we rely on throughout is qualitative and verifiable: the matching layer’s safety properties—now including price-time execution correctness, not only book management—were model-checked at bounded scope with zero invariant violations.

Two boundaries of this provenance deserve equal precision, because “formally verified” is easy to over-read. First, TLC’s guarantee is exhaustive only within the stated bounds — a handful of orders, two prices, short horizons — so it certifies the matching *design* against its invariants at bounded scope; it does not prove unbounded correctness. Second, the object that is verified is the TLA+ specification, not the C++ source: no refinement proof links the two, and the correspondence between spec and implementation rests on construction and conformance testing. Wherever this paper says “verified engine,” both qualifications are attached. The tie-break divergence of §3.2 — our own source-checked Python replica of the clearing rule

differing from the engine the first time the two were run against the same book — is a live demonstration that the second boundary is not pedantry: correspondence claims that are only source-verified can be wrong. What the provenance does buy, and what our internal-validity argument actually uses, is the elimination of design-level matching errors — the kind that could manufacture spurious gaps or transfers — within the checked scope, on the code path every trade in both arms traverses.

1.3 Preview of findings

We organize the evidence as four experiments on this substrate.

- **Experiment 1 (welfare transfer).** With latency-advantaged HFTs present, we measure a positive HFT rent of \$34.70 (95% CI [26.25, 44.80]) that is a near-exact zero-sum transfer from the market maker and noise traders—the welfare residual closes to machine zero (mean 2.8×10^{-12})—while liquidity gaps rise from 2.84 (95% CI [2.47, 3.26]) in the matched no-HFT baseline to 9.70 (95% CI [9.15, 10.27]) in treatment (non-overlapping CIs); the result holds across all nine cells of a 3×3 sensitivity-by-decay calibration.
- **Experiment 2 (latency scaling).** Rent rises with market-maker staleness at approximately \$10.46 per tick, and the data do not distinguish a linear specification from a segmented (knee/phase-transition) alternative ($\Delta\text{BIC} = -0.17$, inside the range of indifference)—an honest negative result: we find no evidence for a critical threshold, with rent first becoming statistically distinguishable from zero at roughly two ticks.
- **Experiment 3 (HFT count).** As the number of HFTs grows, per-HFT rent collapses—a fitted C/k form with $C \approx \$34.8$ ($R^2 = 0.9999$) describes it well, and individual fill rates fall consistent with $1/k$ (the product of fill rate and k is roughly constant at ≈ 0.290)—yet *total* extracted rent stays flat (mean \$34.4, range \$33.70–\$34.93 across $k = 1 \dots 21$); the marginal social cost of added speed competition therefore does not appear as rising aggregate rent but as rising *fragility*—liquidity gaps grow approximately linearly (fitted $\approx 2.1k + 3.7$, $R^2 = 0.997$), roughly $17 \times$ by 21 HFTs (relative to this experiment’s own $k = 0$ baseline of 2.84 gaps).
- **Experiment 4 (batch-auction remedy).** Switching the verified engine to frequent batch-auction clearing (both arms run the same model-checked matcher, §3.2) recovers market-maker welfare monotonically over intervals ≥ 5 —about 97% of the loss is recovered by an interval of 500 (MM loss \$28.5 $\rightarrow$ \$0.7)—and this MM-recovery signal is the robust one; HFT-rent point estimates also fall, but only the interval-5 arm attains a rent interval that clears zero and the remaining batch-arm CIs bracket zero. An extreme interval of 1 *backfires* (MM loss worsens to \$41.3 with about 137 gaps), so the remedy helps only away from that degenerate corner.

1.4 Contribution statement

This paper makes three contributions. First, it provides, to our knowledge, the first experimental test of the BCS welfare decomposition on a *formally verified* matching engine — verified in the precise, bounded sense of §1.2 — removing design-level matching errors as a plausible source of the observed effects and turning the closed-market welfare identity into an auditable, machine-precision check rather than a modeling assumption. Second, it furnishes a complete welfare accounting that decomposes the arms race into market-maker, noise-trader, and HFT components and shows the pathology is a transfer, not value creation. Third, it characterizes the *shape* of the social cost honestly: the cost of speed competition scales smoothly in latency (no phase

transition), and as competitors multiply it manifests as fragility under a flat aggregate rent rather than as ever-growing extraction—while confirming, with appropriate caution about which signal is robust, that the BCS-proposed batch-auction remedy restores liquidity-provider welfare. We view the negative and nuanced results (linear rather than threshold scaling; fragility rather than rising rent; MM-recovery as the clean remedy signal) as integral to the contribution, not caveats to it.

2. Related Literature

This paper sits at the intersection of five strands of work: the market-design theory of the HFT arms race and its direct measurement, the theory of speed and exchange design, the game theory of predatory liquidity provision, the empirical and physics-of-markets literature on endogenous fragility, and the methodological tradition of agent-based modeling. We synthesize each in turn and then state our differentiator.

The arms race and its remedy. Our central reference is Budish, Cramton, and Shim (2015), who reframe latency competition not as a behavior to be regulated but as a symptom of the continuous limit order book’s serial-processing design, and who propose frequent batch auctions as the corrective. BCS pair a theoretical model with reduced-form empirical evidence (notably the persistence of mechanical arbitrage opportunities in correlated instruments). Our work is, in effect, a controlled experimental instantiation of their model: we build the latency-differentiated observation structure and submission-timing race they describe, reproduce their predicted treatment effects (rent transfer and liquidity degradation), and test their proposed remedy by switching the clearing rule of the very same verified engine. We complement rather than re-derive their theory, supplying a closed welfare decomposition that their reduced-form empirics cannot directly observe. The empirical scale of the race has since been measured directly: Aquilina, Budish, and O’Neill (2022), using message-level exchange data that records failed race attempts alongside successful ones, find latency-arbitrage races occurring roughly once per minute per FTSE 100 symbol, decided by microseconds, and summing to a latency-arbitrage “tax” of about 0.4 basis points of trading volume — with the implication that eliminating the race would cut the market’s cost of liquidity by roughly 17%. Their measurement and our experiment are two views of the same transfer: they infer it from races visible in exchange messages; we observe every leg of it — sniper, maker, noise trader — under a conserved accounting identity, at magnitudes that are internally consistent rather than calibrated to theirs.

Speed and exchange design. Two theoretical strands refine the BCS diagnosis and bear directly on our treatment arms. Menkveld and Zoican (2017) show that exchange speed is not neutral infrastructure: a faster matching engine accelerates snipers and market makers alike, raising the frequency of speed duels, so lower exchange latency can *worsen* liquidity when news arrival is high relative to liquidity-motivated trade. Our degenerate interval-1 cell in Experiment 4 — where the most aggressive clearing cadence amplifies the pathology it was meant to cure — is a discrete-clearing cousin of that comparative static, and Experiment 2 traces its analogue in the market maker’s own staleness. Baldauf and Mollner (2020) embed sniping in a model with costly information acquisition and compare design responses — frequent batch auctions and asymmetric speed bumps that delay marketable orders but not cancellations — showing that both blunt the sniping externality, with different side effects on price discovery. Our apparatus tests the first of those designs experimentally; the second is a natural extension, implementable entirely in the latency scheduler without touching the verified matcher.

Predatory versus cooperative liquidity. Carlin, Lobo, and Viswanathan (2007) provide the strategic micro-foundation for why liquidity can evaporate endogenously: when cooperation among strategic traders breaks down, agents race to trade ahead of a distressed counterparty, withdraw liquidity, and amplify price impact, producing episodic crises that owe nothing to exogenous fundamental shocks. The BCS sniping mechanism is, in this light, a latency-mediated modern instance of the predatory equilibrium they formalize. We build on their insight by showing the predatory outcome emerging mechanically from rational speed competition on a concrete book, and by attaching a quantitative welfare price tag to it; we differ in implementing an explicit agent-based experiment with a closed accounting identity rather than a stylized analytic game.

Endogenous fragility and flash crashes. Kirilenko, Kyle, Samadi, and Tuzun (2017) dissect the May 2010 Flash Crash from audit-trail data and find that HFTs, while not the trigger, consumed liquidity and passed inventory among themselves at the critical moment, accelerating the decline. This is the canonical fragility episode our flash-crash diagnostics are motivated by, and our Experiment 3 speaks directly to it by asking whether such fragility can arise from rational HFT behavior alone, with no exogenous shock; we find that it grows with the number of competing HFTs. Where Kirilenko et al. perform forensics on a single historical event, we offer a fully observable, reproducible counterfactual in which HFT count, latency, and market design are experimental knobs. Complementing this, Filimonov and Sornette (2012) introduce the branching ratio of a self-exciting Hawkes process as a measure of market endogeneity (reflexivity), reporting that it spikes toward criticality around fragile periods. We adopt their branching ratio as an independent diagnostic—estimated on our simulated order flow via a Hawkes bridge—to test whether the crashes we model are genuinely self-excited rather than fundamental-driven; the advantage of our setting is that the data-generating process is known and controllable, so the branching ratio can be tied causally to HFT count and latency rather than inferred post hoc from market data of unknown structure.

Agent-based modeling and its verifiability gap. Methodologically, our work extends the agent-based computational finance tradition (Farmer and Foley, 2009; LeBaron, 2006), which argues that bottom-up simulation of heterogeneous, boundedly rational agents can generate emergent, out-of-equilibrium phenomena—fat tails, volatility clustering, bubbles and crashes—that representative-agent equilibrium models cannot. That tradition’s well-known weaknesses are sensitivity to ad hoc behavioral assumptions, weak calibration discipline, and limited reproducibility or verifiability of the simulation machinery itself. We inherit the tradition’s strength (emergent crash dynamics from rational behavior) while directly confronting its central weakness: rather than building on a bespoke, unverified simulator whose matching logic could itself generate the observed artifacts, we drive a TLA+-verified C++ matching engine, and we impose calibration and welfare-accounting discipline in the form of a closed, near-zero welfare residual.

Differentiator. The combination that distinguishes this paper is, to our knowledge, the first experimental test of the BCS welfare decomposition on a formally verified matching engine. BCS is theory plus reduced-form empirics, since joined by the direct message-level measurement of Aquilina, Budish, and O’Neill (2022); the agent-based tradition typically runs on unverified simulators whose matching layer is a candidate source of the very effects under study. Here the entire BCS mechanism lives in a Python scheduler atop a deterministic matcher whose continuous-matching specification was model-checked with zero invariant violations (per the engine’s TLC verification log, `docs/Verification.md`), reached through a pybind11 bridge, so emergent arms-race dynamics, liquidity gaps, and crash behavior cannot plausibly be artifacts of the matching design (in the bounded sense of §1.2). On that substrate we decompose welfare into market-maker, noise-trader, and HFT

components, show the arms race is a transfer that closes to a machine-precision zero-sum residual, trace the mechanism across latency and HFT-count sweeps (with the honest findings that scaling is linear rather than threshold and that aggregate rent is flat while fragility rises), and confirm the BCS remedy experimentally via the batch-auction treatment, with a placebo-validated endogeneity diagnostic in the Filimonov–Sornette spirit that returns an honest null at our operating point. It is the joint presence of a verified engine, a closed welfare identity, and an experimental confirmation of both the pathology and its proposed cure that situates this paper relative to the literature it builds on.

3. Model and Experimental Design

This section describes the simulation apparatus. The design rests on a deliberate separation of concerns: a single formally-verified C++ matching engine adjudicates every trade — continuous matching and batch-auction uncrossing alike — while a Python layer above it supplies the agents, the latency model, and the batch-auction cadence. The verified engine is the part we trust to be correct; the Python layer is the part we vary. The boundary between them is the pybind11 bridge, and it is drawn precisely where the engine’s formal guarantees end.

3.1 The Matching Engine (C++20, TLA+-verified)

All trades in the continuous arm are produced by one C++ matching engine, `OrderMatcher::MatchingEngine`, the same object that is the subject of the project’s TLA+ specification suite. In the BCS harness the engine is run in **synchronous** mode: `EngineHarness::start()` calls `engine_.start()` rather than `startAsync()`, so `submitOrder()` dispatches matching on the calling thread and the `OrderBook`’s `EventListener` fires inline. Draining trades immediately after a submit therefore returns exactly the fills that order produced, which keeps the harness deterministic and free of cross-thread or GIL concerns (`bcs_research/bindings/engine_bindings.cpp`, lines 5–16). The engine offers price-time priority matching, best bid/ask, depth snapshots, VWAP, and an optional circuit breaker; the breaker is disabled in this harness via `setCircuitBreakerThreshold(1e18)` when `start(disable_circuit_breaker=True)` so that simulation control, not an exchange-level halt, governs the dynamics (lines 81–83). Prices are fixed-point: a quoted price of 100.00 is `100 * PRICE_PRECISION`, with `PRICE_PRECISION` re-exported as `bcs_engine.PRICE_PRECISION` (line 186).

Critically, **the engine has no notion of agent latency**. The comment at `engine_bindings.cpp` lines 15–16 states this explicitly: the entire BCS latency/race mechanism lives in the Python scheduler above this layer. The engine’s job is only to be a correct continuous double-auction matcher, and that correctness is what the spec suite certifies.

The directly relevant specification is `Auction.tla` (`spec/Auction.tla`), which models the engine’s session lifecycle over the seven legal `TradingStates` (`PreOpen`, `AuctionOpen`, `Continuous`, `AuctionClose`, `PostClose`, `Halted`, `VolatilityAuction`) and the discrete uncross that is the only way trades emerge from an auction state. Under `Auction.cfg` (`BROKEN_NO_REOPEN=FALSE`, `MaxOrders=3`, `MaxEvents=6`, `Prices={100,200}`), TLC checks the following **invariants**:

- **TypeOK** — the type invariant.

- **ValidState** — every reachable state is one of the seven legal `TradingState`s.
- **NoMatchOutsideContinuousExceptUncross** — no trade prints in an auction, halt, or closed state except a discrete uncross whose source is a true auction state (`PreOpen`, `AuctionClose`, or `VolatilityAuction`). Each trade carries an `inState` history field, which gives the invariant teeth.
- **SingleClearingPrice** — every uncross firing produces trades at exactly one price drawn from `Prices`. This is the code path the batch arm now clears through via the bridge's `uncross` (§3.2).
- **NoTradeWhileHalted** — no trade record may carry `inState=Halted`.

As temporal/action **properties** it checks **ValidTransitions** (`[] [ state'=state \ / <<state,state'>> \ in LegalEdges ]_state` — every step is either a self-loop or a legal directed edge of the session state machine) and the **liveness** property **VolatilityEventuallyReopens** (`[] (state=VolatilityAuction => <> (state \ in {Continuous,Halted}))`), proved under weak fairness `WF_vars(ReopeningUncross)`. Non-vacuity is established by a companion `AuctionBroken.cfg` (`BROKEN_NO_REOPEN=TRUE`) that disables the reopening uncross so TLC reports a liveness counterexample, confirming the property is not vacuously true.

We note honestly that **we did not recover a model-checked state count for `Auction.tla`**: a `states/` directory and TLC trace (`.bin`) artifacts exist in the spec tree, but we did not parse them, and we decline to report a state count we cannot verify. The large reachable-state count cited elsewhere for the engine (§1.2) pertains to `MatchingEngine.tla` and is recorded in `docs/Verification.md`, not derived from `Auction.tla`.

`Auction.tla` sits within a broader suite (`spec/`) that verifies the surrounding engine and infrastructure. We summarize the checked properties rather than restate every configuration:

Spec	Checked properties (per .cfg)
MatchingEngine.tla	NoNegativeQuantity , FIFO_Preservation , GTD_Expiry_Correctness , plus the continuous cross action's FIFOExecution and MatchingConservation (MaxOrders=3 , Participants={1,2} , Prices={100,200} , MaxQty=3 , MaxTime=2 ; a companion MatchingEngineBroken.cfg allows non-FIFO fills so TLC reports a FIFOExecution counterexample, establishing non-vacuity)
FixSession.tla	TypeOK , DeliveredInOrder (BROKEN_NO_GAP_DETECT=FALSE ; constraint BoundedDelivered , MaxSeq=3)
Risk.tla	TypeOK , FirmWithinCap , TradersWithinCap , FirmIsSumOfTraders (BROKEN_NO_FIRM_AGG=FALSE non- vacuity switch)
Oco.tla	TypeOK , ValidStates , AtMostOneWinnerPerGroup , WinnerSilencesSiblings (BROKEN_NO_OCO_CANCEL=FALSE)
Snapshot.tla	TypeOK , SnapshotEntriesAreReal verified; SnapshotIsPointInTime and NoDoubleSidedSnapshot are documented anti-properties that fail under lockless reads (enabled one at a time to produce torn-read counterexamples); companion SnapshotLocked.tla
Replication.tla	NoCommittedLoss , NoDuplicateExecution , NoSplitBrain (lease-based replication; MaxEntries=6 , HeartbeatTimeout=2 , LeaseTimeout=5)
EpochDurability.tla	TypeOK , NoRegression , DurableTracksMax (BROKEN_NO_PERSIST=FALSE resets the epoch on restart so TLC finds the regression; MaxEpoch=3)
MpscQueue.tla	TypeOK , NoSpuriousItems , NoDuplicates , PerProducerFIFO ; liveness EventuallyDrained (under weak fairness on every action)
EngineConsumer.tla	TypeOK , NoSpuriousItems , NoDuplicates ; liveness EventuallyDrained (leads-to: every item placed in any producer queue eventually reaches the consumer log)

The payoff is that any artifact the simulation reports — a fill, a price, a snapshot, a zero-sum residual — is produced by a matcher whose price-time priority, FIFO preservation, non-negativity, and single-clearing-price behaviour have been model-checked, not merely tested.

3.2 The pybind11 Bridge

A pybind11 module named `bcs_engine` bridges the verified C++ engine to Python (`engine_bindings.cpp`, `PYBIND11_MODULE` at line 212). It exposes a single entry class, `EngineHarness`, with: `add_symbol` (which must precede `start()`; `start()` freezes the book registry, after which `add_symbol` throws/no-ops, lines 64–70), `start(disable_circuit_breaker=True)`, `stop`, `submit_order(symbol, order_id, participant_id, side, price, quantity, order_type=Limit, tif=GTC)`, `submit_cancel`, `best_bid`, `best_ask` (with `INT64_MAX` normalized to 0 when the ask side is empty, lines 129–134), `mid` (0 if either side is empty), `drain_trades`, `snapshot(depth=10)`, `vwap`, `trade_count`, and — added to close the matcher seam described below — `uncross` (one uniform-price call-auction clearing of the resting book) together with `set_trading_state/trading_state` over the session states of `Auction.tla` (lines 163–189). Fills are captured by a per-symbol `TradeCollector` that buffers every trade on the book and clears on `drain_trades`; Python attributes each fill to agents by `buyer_id/seller_id`. The enums `Side`, `OrderType` (`Limit/Market/IOC/FOK/PostOnly`), `TimeInForce` (`GTC/GTD/DAY`), and `RejectReason`, plus read-only `Trade`, `PriceLevel`, and `MarketDataSnapshot` classes, are exposed alongside.

Python agents drive the verified engine entirely through this surface: each tick they emit orders and cancels, the scheduler submits them via `submit_order/submit_cancel`, and because the engine is synchronous, the scheduler drains the resulting fills inline and routes them back to the agents. The agents never touch C++ state directly; the bridge is the only channel, which is what lets the verified matcher remain the sole adjudicator of trades.

An earlier revision of this harness carried a methodological seam here: the bridge did not expose `uncross`, so the frequent batch (call) auction could not be run through the engine at all and was instead re-implemented in Python (`discover_uncross_price`), replicating `OrderBook::discoverUncrossPrice` by source inspection. That seam is now closed. The bridge exposes `uncross`, `set_trading_state`, and `trading_state`; the batch arm (`BatchAuctionScheduler`) parks the book in `AuctionOpen` — where orders accumulate instead of matching on arrival — collects the window's orders, and clears through the engine's own `uncross`, the code path `Auction.tla` proves `SingleClearingPrice` over. Both arms of Experiment 4 therefore run on the same verified matcher, and the batch arm's fills are engine fills, drained through the same `TradeCollector` as continuous fills.

Closing the seam produced a finding worth reporting in its own right. The retired Python re-implementation had been checked against the C++ source line by line, and an earlier draft of this paper described it as replicating the rule “faithfully” — a claim we noted was source-verified rather than results-verified. When the two implementations could finally be run against the same book, they disagreed. On a book where two prices tie on maximum executable volume (bids 102×50 and 101×30 against asks 99×40 and 100×60 , so 80 units clear at either 100 or 101), the Python rule clears at 100 and the verified engine clears at 101 — a divergence inside the tie-break cascade, invisible to any volume-based check and material to every metric that depends on execution price, rent above all. The divergence is pinned in the test suite (`tests/test_batch_auction.py::test_python_matcher_divergence_pinned`) so it cannot regress silently, and `discover_uncross_price` is retained only as a cross-check. The episode is a compact argument for this paper's central methodological claim: a re-implementation that is source-verified can still differ from the verified artifact the first time it is results-verified.

3.3 Agents

All agents are Python objects driven by the scheduler. They share a common interface — `act(t, market, fundamental)` returns a list of `Action`s, and `on_fill(trade, is_buyer)` receives routed fills — and each holds its own cash and inventory for the welfare accounting of §3.5.

FundamentalValueProcess (`bcs_research/agents/fundamental_value.py`) is the exogenous true value $V(t)$, a discrete random walk in fixed-point price space. On `step(t)` it draws $dv = \sigma * N(0,1)$ (per-step volatility, independent of tick size; `dt_us` is wall-clock metadata only), updates `current_v += dv`, and records `(t, current_v)`. Its `observe(t, latency_us)` returns the most recent recorded V at or before `t - latency_us` via bisection — this latency-delayed read is the BCS information gap that distinguishes a fast agent from a slow one. The undelayed `current_v` exposes the true value used for the market maker's adverse-fill check. The initial value is `v0 = 100 * PRICE_PRECISION` in Experiment 1, and a NumPy `default_rng(seed)` provides determinism.

BCSMarketMaker (`bcs_research/agents/bcs_market_maker.py`, `participant_id = MM_ID = 1`) is an adverse-selection-aware two-sided quoter. On each requote tick (every `reqoute_every` ticks) it first decays its half-spread toward the base — `current_half_spread = current_half_spread * (1 - spread_decay) + base_half_spread * spread_decay` — then observes V with its larger latency (`latency_us = 3000` us in Experiment 1, so its quotes are stale), and posts `bid = round(v - hs - skew)`, `ask = round(v + hs - skew)` with `skew = inventory * inventory_skew`. It emits a cancel of the old bid/ask plus new two-sided Limit orders, all stamped `submit_at = t + latency_us`. Two coupled defences drive the mechanism:

- *Adverse-only widening* (`on_fill`): the maker widens only when a fill is against informed flow — it bought *above* or sold *below* the true `current_v` by more than `adverse_threshold_ticks`. Noise fills, which sit inside `[bid, ask]` around V , do not widen. On an adverse fill it increments `pickoffs` and sets `current_half_spread = min(current_half_spread * (1 + adverse_sensitivity), spread_cap)` with `spread_cap = base * spread_cap_mult`.
- *Liquidity thins as it widens* (`_effective_qty`): resting size is `max(min_quote_qty, round(quote_qty * base_half_spread / current_half_spread))`, so size shrinks toward `min_quote_qty` as the half-spread grows. Spread and depth are one coupled state, not two.

It also tracks `max_half_spread_seen`, `min_quote_qty_seen`, `inventory`, and `cash`. Calibration of `adverse_sensitivity` and `spread_decay` (so the no-HFT spread sits near base) is done by the experiment driver, not the agent class.

HFTAgent (`bcs_research/agents/hft_agent.py`, `participant_id = 200 + j`) is a latency-advantaged stale-quote sniper. Each tick it observes V with a small latency (`latency_us = 0` in Experiment 1, so it sees V immediately) and reads `best_bid / best_ask`. It fires a buy snipe — an aggressive IOC Buy at `best_ask` — when `(v - best_ask) > transaction_cost * best_ask`, and a sell snipe — IOC Sell at `best_bid` — when `(best_bid - v) > transaction_cost * best_bid`, where `transaction_cost = transaction_cost_bps / 10000`. Each acted opportunity increments `snipes`. The race is a random

submission jitter `int(uniform(0, race_noise_us))` added to `submit_at (= t + latency_us + jitter)`: with several HFTs the lowest jitter wins and the others' IOCs find the quote already gone. The counterparty is the larger-latency maker; that latency gap is the BCS arbitrage.

NoiseTrader (`bcs_research/agents/noise_trader.py`, `participant_id = 100 + i`) is the uninformed liquidity taker and the control population. Each tick, with probability `lambda_per_tick`, it crosses the spread with an IOC order on a random side (50/50): buying at `best_ask` or selling at `best_bid` (and returning nothing if that side is empty), with `submit_at = t + latency_us`. It carries no fundamental-value signal and never snipes; it pays the half-spread to the maker, so its expected PnL is negative by construction. It is the counterfactual flow against which informed (HFT) flow is measured.

3.4 The BCS Mechanism

The Budish–Cramton–Shim (BCS) arms-race feedback loop is not a single equation but a closed cycle realized across the agents and the continuous scheduler (`bcs_market_maker.py`, `hft_agent.py`, `scheduler.py`):

1. **V moves.** The HFT sees the new `V` with small latency (0 us in Experiment 1) while the `BCSMarketMaker` still has quotes posted around its latency-delayed view of `V` (3000 us). The maker's quote is now **stale**.
2. **Snipe.** The HFT detects `(v - best_ask) > cost` or `(best_bid - v) > cost` and fires an aggressive IOC to pick off the stale quote, winning the submit-time race among HFTs via the lowest race jitter.
3. **Widen.** The maker's `on_fill` recognizes the fill as informed (bought above / sold below the true `current_v`) — an adverse fill — and widens its half-spread multiplicatively (`current_half_spread *= 1 + adverse_sensitivity`, capped at `base * spread_cap_mult`), incrementing `pickoffs`.
4. **Thin.** As the half-spread widens, the maker's effective resting size shrinks toward `min_quote_qty` (`qty = quote_qty * base / current_half_spread`).
5. **Fragility.** A thinner book means the next order moves the mid further or empties a level — larger price impact — so more quotes go stale and the cycle repeats: snipe → widen + thin → larger impact → more stale quotes → more sniping. The half-spread decays back toward base each requote tick, so the system recovers when not being picked off.

The **LatencyScheduler** (`bcs_research/simulation/scheduler.py`) is the race substrate that supplies the latency model the C++ engine lacks. Each tick (advancing `dt_us`) it steps the fundamental, resets per-tick volume and signed volume, lets every agent emit `Actions`, and enqueues them in a heap keyed (`submit_at`, `seq`). Then `_submit_due(t)` pops and submits every action with `submit_at <= t` in submit-time order (cancels via `submit_cancel`, orders via `submit_order`), draining trades after each submit and routing fills to `buyer.on_fill(..., True)` / `seller.on_fill(..., False)` by `participant_id`. The BCS submission race is exactly many agents stamping near-identical `submit_at` values, resolved here in time order through the heap — there is no special-case race logic, only priority over arrival time. The scheduler records per-tick snapshots (`t`, `best_bid`, `best_ask`, `mid`, `v`, `tick_volume`, `signed_volume`) and per-fill trades into a `SimResult`, with tick quantization at `dt_us`.

The endogenous-instability signature of this specific mechanism is **not a mid dislocation but a liquidity gap**: a transient one-sided or empty book, where a thinned level is emptied before the maker requotes (`liquidity_gap_detector.py`, module docstring, lines 1–9). An event qualifies as an endogenous liquidity gap only if it is a maximal run of non-two-sided snapshots (`best_bid <= 0` or `best_ask <= 0`) that (a) **recovers** to a two-sided book before the run ends and has `duration_us = gap_ticks * dt_us <= max_gap_us` (50,000 us; otherwise it is a structural outage and is discarded), and (b) is **endogenous**, i.e. V is flat across it: `abs(V_hi - V_lo) < k_fund * sigma_per_step * sqrt(gap_ticks + 2)` with `k_fund = 3.0` and one tick of margin on each side (lines 36–43). Value-driven gaps — where V moves above the threshold — are excluded. We chose this event definition because an earlier all-fills version of the maker (which widened on every fill, not just adverse ones) produced no HFT-specific divergence in mid prices; widening is therefore gated on adverse fills, and the instability surfaces as gaps in depth rather than crashes in price.

3.5 Metrics

All metrics are pure functions over a `SimResult` and apply unchanged to both the continuous and batch arms, since `BatchAuctionScheduler` emits the same `SimResult` shape.

Welfare decomposition and the zero-sum identity (`bcs_research/metrics/welfare_decomposition.py`). `decompose_welfare(result, mm_id, nt_ids, hft_ids, mark)` computes per-agent PnL as `cash + inventory * mark / PRICE_PRECISION`, marked at `mark_price` (the last two-sided mid, else the last V). It reports `mm_pnl`, `noise_trader_pnl` (summed over `nt_ids`), `hft_pnl`, `hft_rent` (equal to `hft_pnl` — the rent the arms race extracts), per-agent and per-second variants, and `zero_sum_residual = mm_pnl + nt_pnl + hft_pnl`. In a closed market the residual is approximately zero — the conservation check that **closes through the verified engine**: because every fill is produced by the model-checked matcher with FIFO preservation and non-negative quantities, cash and inventory are conserved exactly, and any non-trivial residual would indict the harness rather than the market. `welfare_delta` computes the treatment-minus-baseline difference for the key terms; this welfare transfer is the paper’s primary treatment effect, because it moves even when quoted spreads do not.

Lagged Kyle’s lambda (`bcs_research/metrics/baseline_metrics.py`, `compute_metrics`). `kyle_lambda` is the OLS slope (`np.polyfit` degree 1) of the *forward-window* mid change `mid[t+k] - mid[t]` (default `k = kyle_horizon_ticks = 5`) regressed on `signed_volume` at t , over ticks with `mid > 0`; it requires at least 3 points and `std(x) > 0`, and `kyle_lambda_r2` is the squared correlation. The forward lag, rather than the contemporaneous tick, is deliberate and documented (lines 64–70): a latency-disadvantaged maker quotes stale, so a same-tick regression would score its delayed catch-up requote as *negative* impact; measuring over the next k ticks captures the catch-up as the *positive* permanent impact it actually is, removing the stale-quote sign artifact.

Liquidity-gap detector (`bcs_research/metrics/liquidity_gap_detector.py`). As defined in §3.4, `detect_liquidity_gaps` returns the bounded, recovering, endogenous one-sided episodes; `count_liquidity_gaps` returns the count. This is the secondary endogenous-instability metric, matched to the thinning mechanism (gaps, not mid crashes), and each event records `start_t`, `recovery_t`, `duration_us`, `gap_ticks`, `side_missing`, and `v_move`.

Percentile bootstrap CI (`bcs_research/metrics/bootstrap.py`). `bootstrap_ci(values, n_bootstrap=2000, ci=0.95, seed=0)` is a nonparametric percentile bootstrap for the mean: it draws `(n_bootstrap, n)` resamples with replacement via `np.random.default_rng(seed)`, takes per-row means, and returns the (seed-independent) point estimate `mean`, the lower / upper bounds at the `(alpha/2, 1 - alpha/2)` percentiles, `std` (`ddof=0`), and `n`. A single value yields a zero-width CI. `bootstrap_ci_table` builds a `{metric_key: ci}` table from per-seed row dicts. The resampling RNG is explicitly seeded for reproducibility. The `n_bootstrap=2000` resample count and percentile method are defaults of the bootstrap module (source-level), not fields stored in the results JSON, which records only the bounds and `n`. We note that `exp1_primary.py` itself does not call `bootstrap_ci` — it aggregates per-seed rows with `statistics.fmean` — and the bootstrap module is shared infrastructure used from Experiment 2 onward; the `n=100` CIs the paper reports for Experiment 1 come from the hardened run, which does use the bootstrap.

3.6 Experimental Design Summary

The primary operating point is the `CFG` dict in `bcs_research/experiments/exp1_primary.py`. Both arms share identical market parameters and differ **only in HFT presence**: the **baseline** is the `BCSMarketMaker` (latency-disadvantaged, adverse-selection-aware) plus noise traders; the **treatment** is the baseline plus `N` `HFTAgents`. Both run on the continuous `LatencyScheduler` through the verified engine. The headline configuration is:

Parameter	Value
<code>duration_us / dt_us</code>	3,000,000 / 1,000 (3,000 ticks)
Noise traders / HFTs	8 / 3
<code>base_half_spread / quote_qty</code>	50 ticks (full spread 100 ticks = \$0.01) / 50
<code>sigma</code> (per step)	20.0
<code>mm_latency_us</code>	3,000 (stale quotes)
<code>mm_inventory_skew</code>	0.2
<code>adverse_sensitivity / spread_decay / spread_cap_mult / min_quote_qty</code>	0.1 / 0.1 / 10.0 / 1
<code>hft_latency_us / hft_race_noise_us / hft_qty / hft_cost_bps</code>	0 / 100 / 50 / 0.0
<code>lambda_per_tick / noise_qty</code>	0.15 / 10
<code>v0</code> , symbol, MM id, noise ids, HFT ids	<code>100 * PRICE_PRECISION</code> , 1, 1, <code>100 + i</code> , <code>200 + j</code>
Circuit breaker	disabled
Seeds	20 (per-seed metrics averaged by <code>fmean</code>); reported CIs via 2,000-resample percentile bootstrap (module default), 95%

Per-seed outputs (`mm_pnl`, `nt_pnl`, `hft_rent` and `_per_sec`, `zero_sum_residual`, `liquidity_gaps`, `kyle_lambda`, `spread_ticks`, `n_trades`, `mm_max_half_spread`, `mm_min_qty`, `hft_snipes`, `hft_fills`) are written to `bcs_research/results/experiments/exp1_primary.json`; the hardened `n = 100` results

reported in §4.1 come from `exp1_hardening.py` and are written to `exp1_hardening.json`.

Around this operating point we run four one-dimensional sweeps, each isolating one axis of the mechanism:

1. **Calibration robustness (3×3).** `adverse_sensitivity` × `spread_decay` over `{0.05, 0.10, 0.20}`, confirming the result is not an artifact of the calibrated cell — HFT rent must be significantly positive and the zero-sum identity must hold in all nine cells.
2. **Latency sweep (Exp 2).** The maker's staleness gap from 0 to 20 ticks with `hft_latency = 0`, on a tick-aligned grid (latency resolves only at `dt_us` granularity), testing whether rent rises linearly or exhibits a knee/phase transition.
3. **HFT count sweep (Exp 3).** Number of HFTs over `{0, 1, 2, 3, 5, 8, 13, 21}`, separating per-HFT rent (competed away) from total rent (transfer) and from fragility (liquidity gaps).
4. **Auction-frequency sweep (Exp 4).** Batch interval over `{1, 5, 10, 25, 50, 100, 500}` plus continuous, run through `BatchAuctionScheduler` (which parks the book in an auction state and clears through the engine's own `uncross`, §3.2), measuring how much maker welfare loss the frequent batch auction recovers and at what interval.

Each sweep varies one parameter and holds the rest at the operating point, so that any change in the outcome metrics is attributable to that single axis. We note that the no-HFT baseline liquidity-gap figure is reported per experiment from its own matched run. At $n = 100$ these agree: the matched baseline anchoring the Experiment 1 headline comparison is 2.84 gaps (95% CI [2.47, 3.26]), and the Experiment 3 $k = 0$ cell that anchors the fragility ratio reports the same 2.84. An earlier draft carried a discrepancy here — an $n = 20$ baseline of 3.45 against an $n = 100$ figure of 2.84 — which we attributed to sample size; raising every experiment to $n = 100$ resolves it, confirming that reading.

3.7 Calibration to BTC order flow

To discipline the operating point against real microstructure, we calibrate the sim's *environment* parameters to Binance BTCUSDT-perpetual order flow recorded by a companion collector (20-level book snapshots at ~ 10 /s and the aggregated-trade stream, both with nanosecond timestamps). The collector drops a few hours on most days (host sleep), which we handle by making each moment gap-safe rather than by discarding gapped days: arrival intensity accumulates each hourly file's own observed span, so missing hours leave the denominator instead of inflating it, and realized volatility keeps only returns between *adjacent* one-second buckets, so no return straddles a gap. Days enter at 18 hours of coverage in both streams; files truncated by a mid-write crash are skipped and recorded. This is not housekeeping: a whole-day span over a 21/24 day biases the arrival rate down by roughly 12%, and demanding 24/24 coverage admits only two of the days the collector has ever produced. Under the gap-safe estimators the sample is 27 days (2026-06-24 to 2026-07-21, less 2026-07-19; 37.0M trades, 4.16M snapshots, 2.27M seconds of coverage), yielding: trade arrival 16.28/s; trade sizes p50/p90/p99 of 0.003/0.177/2.01 BTC; median spread of one exchange tick (≈ 0.016 bp); median top-of-book depth 6.18 BTC; and 1-second realized volatility of 0.650 bp/ $\sqrt{s}$ (`results/calibration/targets_btcsd.json`).

The mapping between sim ticks and wall-clock is a modelling choice we fix at 1 tick = 100 ms: the market maker's 3-tick latency then means 300 ms of quote staleness — slow but plausible — while HFTs react within a tick, preserving the BCS fast/slow asymmetry at realistic crypto scales. Under that convention two

environment parameters invert directly, and both fits verify by simulation (`calibration/fit.py`). The volatility inversion is clean: at fitted σ the baseline’s simulated realized vol is 0.647 vs the target 0.650 bp/ $\sqrt{s}$. The arrival inversion is not: its closed form assumes every noise order trades, but only about half the IOCs find a quote at the operating point, so the fit iterates the arrival intensity against the *simulated* trade rate to a fixed point (converging in three steps from a closed-form 0.204 to 15.94 vs 16.28/s). The fitted values are `lambda_per_tick` = 0.427 and σ = 20.6 against the pre-registered operating point’s 0.15 and 20.

The volatility figure is worth dwelling on, because an earlier version of this calibration reported σ = 55.7 — nearly three times the operating point — from the two days that passed a 24/24 completeness filter. Those two days were simply hot (1.76 bp/ $\sqrt{s}$ against the 27-day 0.650), and a spot-checked quiet day ran colder still (2026-07-19, ~ 0.33 bp/ $\sqrt{s}$). Estimated over the full month the fitted σ lands within 3% of the value the experiments were pre-registered at, so on volatility the operating point is not data-adjacent but data-consistent; the earlier discrepancy was a two-day sampling artifact, and we report it here because it is exactly the kind of artifact a small “clean-data-only” sample invites. Arrival intensity remains genuinely under-set: the fitted per-trader rate is $2.8\times$ the operating point’s, i.e. the experiments run on a market thinner in order flow than the perp. Experiments 1–4 are reported at the pre-registered operating point; the calibration locates that point relative to real flow rather than replacing it, and re-running the experimental grid at fitted parameters is future work.

Equally important is what this calibration does *not* claim. Spread and depth levels are not fitted: the stylized MM quotes a ~ 1.5 bp median spread where the perp trades at ~ 0.016 bp, and its depth-to-median-trade ratio is $5\times$ against the perp’s $\sim 2060\times$. The size distribution is not fitted in the experiments of §4: sim order sizes there are constants while real sizes are heavy-tailed ($p_{99}/p_{50} \approx 670\times$). A lognormal size model for the calibrated re-run is implemented and parameterized ($\sigma_{\ln} = 2.80$ fitted on the p_{50}/p_{99} ratio; median 1 unit with a calibrated book depth of ~ 2061 units, matching the real depth-to-median-trade ratio), with the disclosed compromise that a two-quantile lognormal under-predicts the p_{90} size ratio, $36\times$ against the observed $59\times$. And the latency/race structure is not estimable from any public crypto feed — measuring races requires failed-order message data of the kind Aquilina, Budish, and O’Neill (2022) obtained from a regulator — so `hft_latency_us` and the race remain swept design parameters throughout. The calibrated claim is therefore deliberately narrow: the *environment* the agents trade in (arrival intensity, fundamental volatility) is disciplined by data; the strategic structure is designed, not estimated.

4. Results

We report four completed experiments, all run through the same formally verified matching engine: continuous-market welfare decomposition (4.1), latency-advantage scaling (4.2), the HFT competition arms race (4.3), and batch auctions as a remedy (4.4). A preliminary Hawkes-bridge endogeneity probe did not reach significance and is not a main-paper result; we retain it only as supplementary scaffolding for forthcoming work, clearly demarcated in §4.5 below. The cross-cutting validation throughout is the zero-sum conservation residual, which closes to floating-point noise on every run, so that each welfare decomposition is an audited accounting identity rather than a modelling assumption.

4.1 Experiment 1: Welfare decomposition and the zero-sum conservation check

We begin by establishing that the welfare accounting in our setting is an audited identity rather than a modelling assumption. By construction, every unit of profit-and-loss in the formally verified matching engine accrues to one of three agent classes — high-frequency traders (HFT), market makers (MM), or noise traders (NT) — so the sum of realised PnL across classes must equal zero on every run. We measure the empirical conservation residual across 100 seeds at the operating point ($n_{\text{hft}} = 3$, adverse sensitivity 0.1, spread decay 0.1, MM latency $3000\mu\text{s}$). In the treatment economy the residual closes to machine zero, $2.78\text{e-}12$ $[-3.34\text{e-}12, 9.11\text{e-}12]$, and the matched no-HFT baseline behaves identically, $-5.39\text{e-}12$ $[-2.16\text{e-}11, 1.05\text{e-}11]$. Because the residual is indistinguishable from floating-point noise in both arms, the decomposition of welfare into HFT rent, MM loss, and NT loss is a verified accounting identity, and the rent magnitudes reported below are not subject to unmodelled leakage.

Within this conserved frame, HFT entry extracts a positive rent of 34.70 [26.25, 44.80] per run. The transfer is borne almost entirely by market makers: MM PnL changes by -28.50 $[-38.31, -20.35]$ relative to the matched baseline (from 101.27 [93.28, 110.72] to 72.78 [71.05, 74.38]), while the noise-trader class absorbs a much smaller share, with an NT PnL delta of -6.21 $[-7.87, -4.61]$. At $n = 100$ this NT delta is now statistically distinguishable from zero — at $n = 20$ its interval straddled zero — but it remains an order of magnitude smaller than the MM delta and accounts for roughly 18% of the transfer. The welfare transfer is therefore predominantly, though not exclusively, borne by liquidity suppliers, consistent with HFT rent arising chiefly from picking off stale market-maker quotes rather than from worsened execution for uninformed flow.

The microstructure correlates of this transfer are pronounced. Liquidity gaps more than triple under HFT entry, 9.70 [9.15, 10.27] in treatment against 2.84 [2.47, 3.26] in the matched no-HFT baseline, a delta of 6.86 [6.20, 7.52]. Lagged Kyle’s λ , our measure of price impact per unit signed flow, rises by roughly an order of magnitude, from 0.200 [0.184, 0.216] to 1.372 [1.361, 1.383]. Table 1 collects these estimates.

Table 1. Experiment 1 welfare and microstructure estimates at the operating point ($n = 100$ seeds; point estimate with 95% bootstrap CI).

Quantity	Treatment	No-HFT baseline	Treatment – baseline
HFT rent	34.70 [26.25, 44.80]	—	—
MM PnL	72.78 [71.05, 74.38]	101.27 [93.28, 110.72]	-28.50 $[-38.31, -20.35]$
NT PnL	-107.48 $[-116.84, -99.26]$	-101.27 $[-110.72, -93.28]$	-6.21 $[-7.87, -4.61]$
Zero-sum residual	$2.78\text{e-}12$ $[-3.34\text{e-}12, 9.11\text{e-}12]$	$-5.39\text{e-}12$ $[-2.16\text{e-}11, 1.05\text{e-}11]$	—
Liquidity gaps	9.70 [9.15, 10.27]	2.84 [2.47, 3.26]	6.86 [6.20, 7.52]
Lagged Kyle’s λ	1.372 [1.361, 1.383]	0.200 [0.184, 0.216]	—

We stress-test the rent result against the two parameters that most directly govern the adverse-selection channel, sweeping adverse sensitivity and spread decay over the grid $\{0.05, 0.1, 0.2\} \times \{0.05, 0.1, 0.2\}$ with $n = 100$ seeds per cell. HFT rent is significantly greater than zero in all nine cells, and the zero-sum identity holds in all nine, with point estimates of the rent ranging from 16.56 to 46.63. The rent is thus a robust feature of the calibration rather than an artefact of a single operating point.

Two points of nuance should be stated plainly rather than relegated to caveats. First, the no-HFT baseline is mildly fragile: liquidity gaps in that arm are not exactly zero (2.84 [2.47, 3.26]), reflecting residual quote churn even absent HFT participation, so the baseline does not represent a frictionless market. We nonetheless interpret the treatment-versus-baseline contrast as informative because the gap CIs are non-overlapping (treatment 9.70 [9.15, 10.27] versus baseline 2.84 [2.47, 3.26]). Second, the NT PnL delta is small but, at $n = 100$, statistically distinguishable from zero (-6.21 [-7.87, -4.61]); we therefore state that HFT entry imposes a measurable but secondary cost on noise traders, roughly 18% of the total transfer against the market maker's 82%. The welfare story of Experiment 1 remains a predominantly market-maker-to-HFT transfer occurring under a conserved, machine-zero accounting identity, accompanied by a sharp deterioration in measured liquidity.

4.2 Experiment 2: Latency-advantage scaling

Experiment 2 isolates the dependence of high-frequency-trader (HFT) rent extraction on the magnitude of the market-maker's (MM) quote staleness, holding HFT latency fixed at 0 us. The design question is whether the relationship exhibits a *phase transition*—a critical staleness threshold beyond which rent extraction accelerates discontinuously—or whether rent simply scales smoothly with the latency advantage. The result is an honest negative on the phase-transition hypothesis: HFT rent rises approximately linearly in MM staleness, with no detectable knee. We report this negative as a substantive finding, not a caveat.

We sweep MM staleness over an eight-point grid, tick-aligned at the matching-engine resolution $dt_{us} = 1000$, so that latency resolves only at integer multiples of one tick: [0, 1, 2, 3, 5, 8, 13, 20] ticks, equivalently [0, 1000, 2000, 3000, 5000, 8000, 13000, 20000] us. Each grid cell is run over 100 seeds, yielding $n = 800$ fitted observations. The tick-aligned grid is deliberate: because the engine timestamps events at dt_{us} granularity, sub-tick latency offsets are not realizable, and a continuous staleness axis would misrepresent the resolution at which the advantage actually materializes.

Figure 2a plots HFT rent against MM staleness with the linear-fit overlay and the significance threshold annotated.

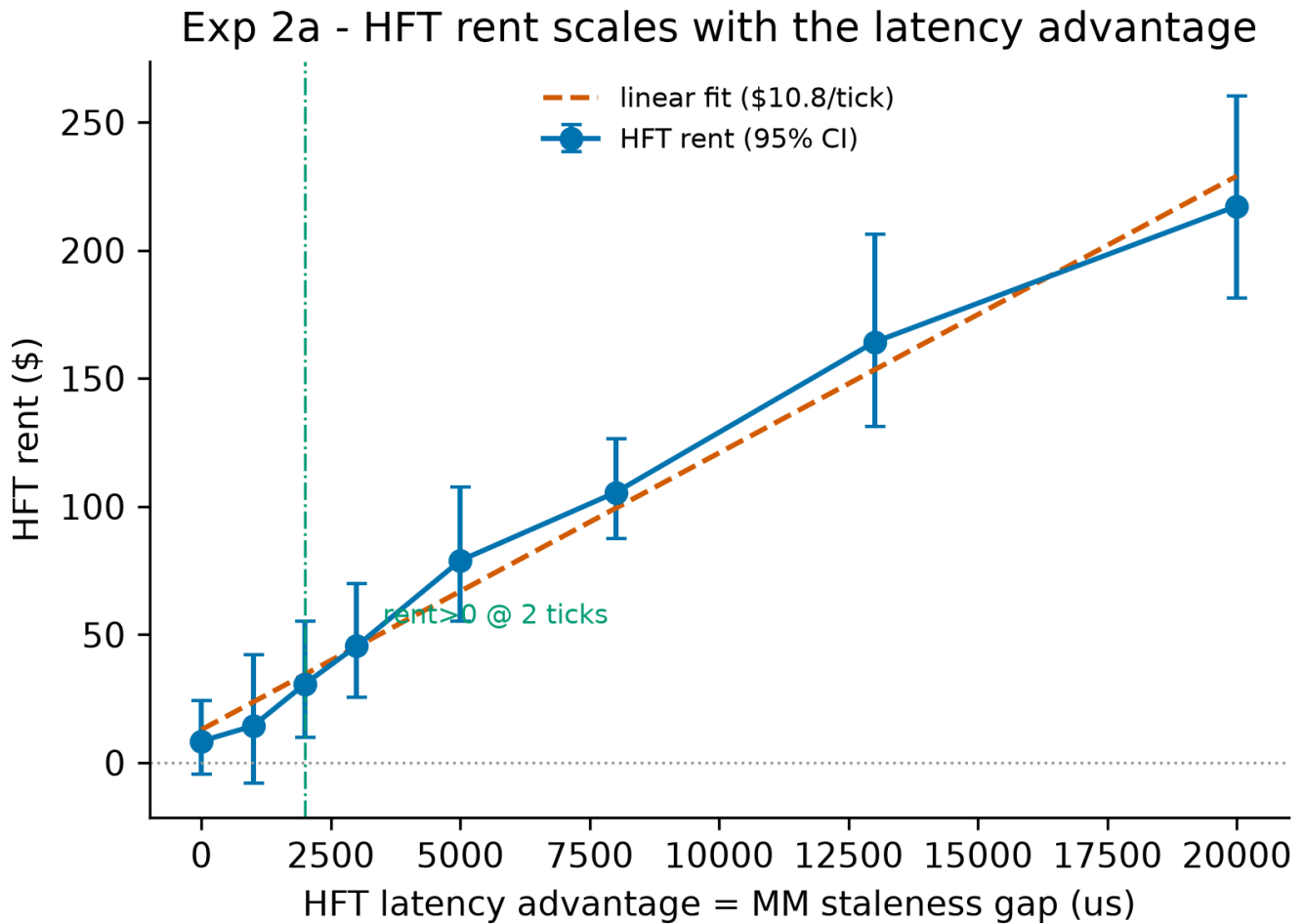


Figure 2a: HFT rent versus MM staleness with linear-fit overlay and significance threshold

The linear fit gives a slope of \$10.460 per tick of MM staleness on an intercept of \$4.710. Formal model comparison is, at this sample size, close to indifferent between the linear specification and a segmented (knee) alternative: ΔBIC (linear minus segmented) = -0.173 . The sign nominally favours the linear model, but a magnitude below 2 is conventionally regarded as “not worth more than a bare mention,” so we do not read it as evidence for linearity — only as an absence of evidence for a break. The pipeline’s recorded verdict remains `linear_scaling`, but that label is the fallback the code emits whenever `strong_evidence` is false (the threshold is $\Delta\text{BIC} \geq 10$); it should be read as “no strong evidence of a phase transition,” not as positive support for a straight line.

We report the segmented fit’s parameters accordingly: a knee at 6510.2 us (≈ 6.5 ticks), a pre-knee slope of \$13.696 per tick, and a post-knee slope of \$8.924 per tick. The evidence does not prefer segmentation (`prefers_segmented`: false) and support for a structural break is not strong (`strong_evidence`: false). The pattern the larger sample reveals is mild concavity — rent accumulates slightly faster at low staleness and slower at high staleness, consistent with HFT fill-rate saturation (the fill rate rises from 0.006 to 0.148 across the grid but flattens markedly beyond 8 ticks) rather than with a critical threshold.

This is a change from our earlier $n = 20$ estimates, and we flag it explicitly. At $n = 20$ the same comparison gave $\Delta\text{BIC} = -7.86$, which we had read as positive support for linear scaling. The larger sample does not overturn the qualitative conclusion — we still find no phase transition — but it does dissolve the statistical basis for the stronger claim. What survives is the negative result: no critical latency threshold is detectable in these data. What does not survive is the affirmative claim that the data prefer a straight line.

Rent is not statistically distinguishable from zero at the smallest staleness levels. The per-cell rent estimates (point [lo, hi], 95% bootstrap CI) are:

MM staleness (ticks)	staleness (us)	HFT rent (\$)	HFT fill rate	Liquidity gaps
0	0	-0.84 [-6.64, 5.92]	0.00601 [0.00539, 0.00663]	0.56 [0.41, 0.73]
1	1000	6.85 [-1.93, 16.79]	0.04349 [0.04267, 0.04432]	3.11 [2.84, 3.39]
2	2000	25.76 [16.58, 35.88]	0.07664 [0.07570, 0.07755]	6.65 [6.25, 7.04]
3	3000	34.70 [26.25, 44.80]	0.09642 [0.09569, 0.09715]	9.70 [9.15, 10.27]
5	5000	66.55 [56.71, 77.79]	0.11607 [0.11537, 0.11679]	17.37 [16.61, 18.18]
8	8000	97.33 [89.13, 106.55]	0.12957 [0.12895, 0.13019]	28.73 [27.66, 29.80]
13	13000	146.39 [136.17, 158.05]	0.14110 [0.14045, 0.14171]	49.14 [47.57, 50.84]
20	20000	204.85 [191.48, 219.23]	0.14787 [0.14716, 0.14863]	75.64 [73.61, 78.03]

The rent CIs at 0 ticks (-0.84 [-6.64, 5.92]) and 1 tick (6.85 [-1.93, 16.79]) both straddle zero; the interval first clears zero at 2 ticks (25.76 [16.58, 35.88]), so we take 2.0 ticks = 2000 us as the significance threshold at which the latency advantage becomes economically detectable. This is a detectability floor, not a phase boundary: once positive, rent continues to grow at a near-constant marginal rate consistent with the \$10.460-per-tick slope, rather than jumping at any subsequent level. We note that the zero-staleness control now centres essentially on zero (-0.84, against 8.44 at $n = 20$), which is the behaviour the design predicts: with no latency advantage there is no rent to extract, and the earlier positive point estimate was sampling noise.

Figure 2b shows the zero-sum welfare decomposition against staleness.

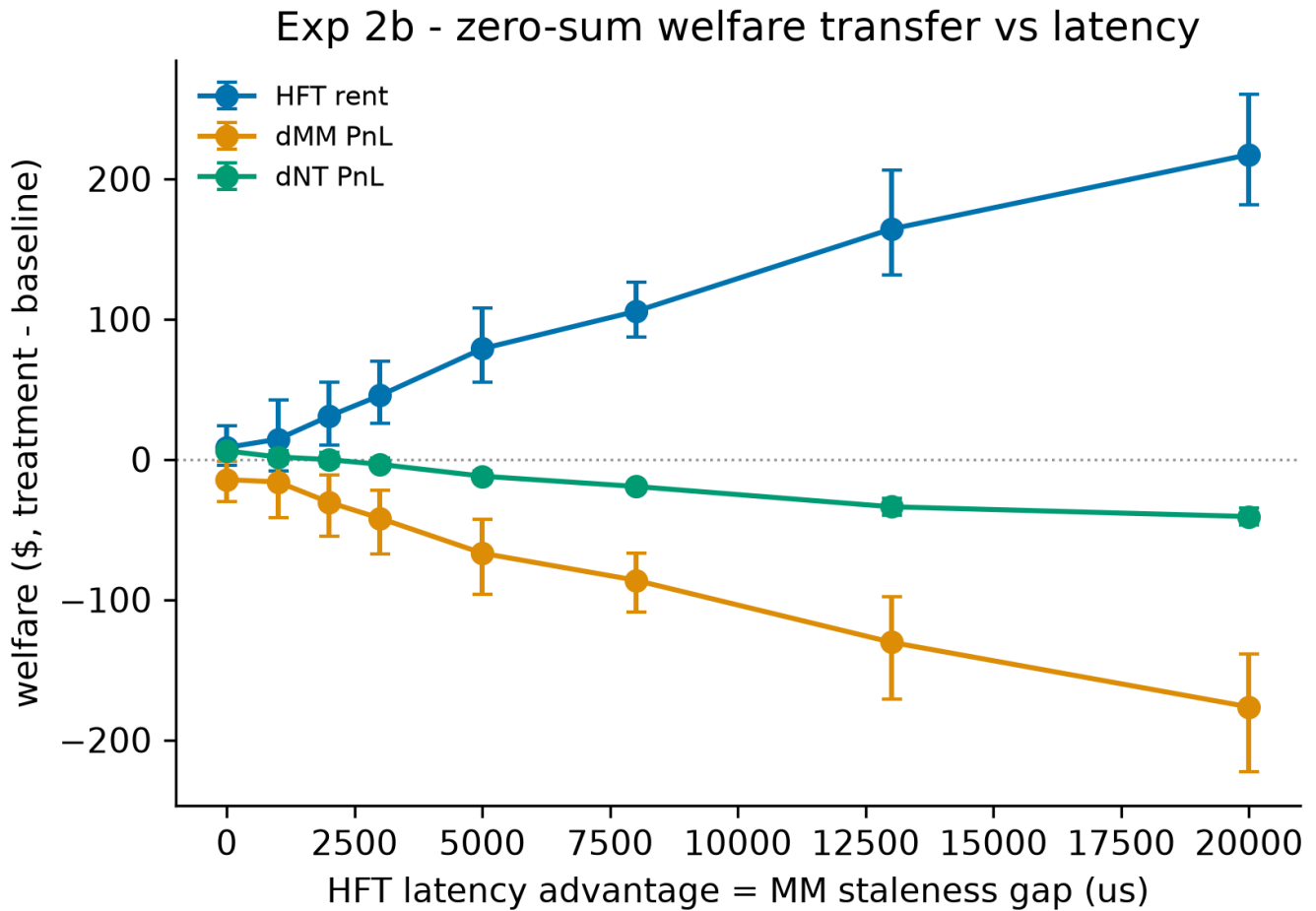


Figure 2b: Zero-sum welfare decomposition versus MM staleness

HFT rent is financed by the other participants. Treatment-arm MM PnL declines monotonically from 93.21 [89.67, 96.83] at 0 ticks to -53.20 [-60.63 , -45.97] at 20 ticks, with the point estimate crossing into losses by 13 ticks (-9.35 [-14.08 , -4.72]); at $n = 100$ that 13-tick loss is now statistically significant, whereas at $n = 20$ its interval still included zero. Noise-trader PnL deteriorates from -92.37 [-101.49 , -84.50] to -151.66 [-161.58 , -143.05] over the same range. The decomposition is clean: the zero-sum residual is ~ 0 across all cells ($\max |\text{mean}| \approx 1.5 \times 10^{-11}$), confirming that the gains and losses net out to a pure welfare transfer with no leakage. The price-impact channel is consistent with this story—treatment Kyle’s λ rises from 0.1885 [0.1743, 0.2024] at 0 ticks, peaks at 1.3716 [1.3605, 1.3826] at 3 ticks, and relaxes to 0.3762 [0.3667, 0.3860] by 20 ticks—indicating that the marginal informativeness of order flow is concentrated at intermediate staleness rather than at the extremes.

Figure 2c reports liquidity gaps alongside the HFT fill rate.

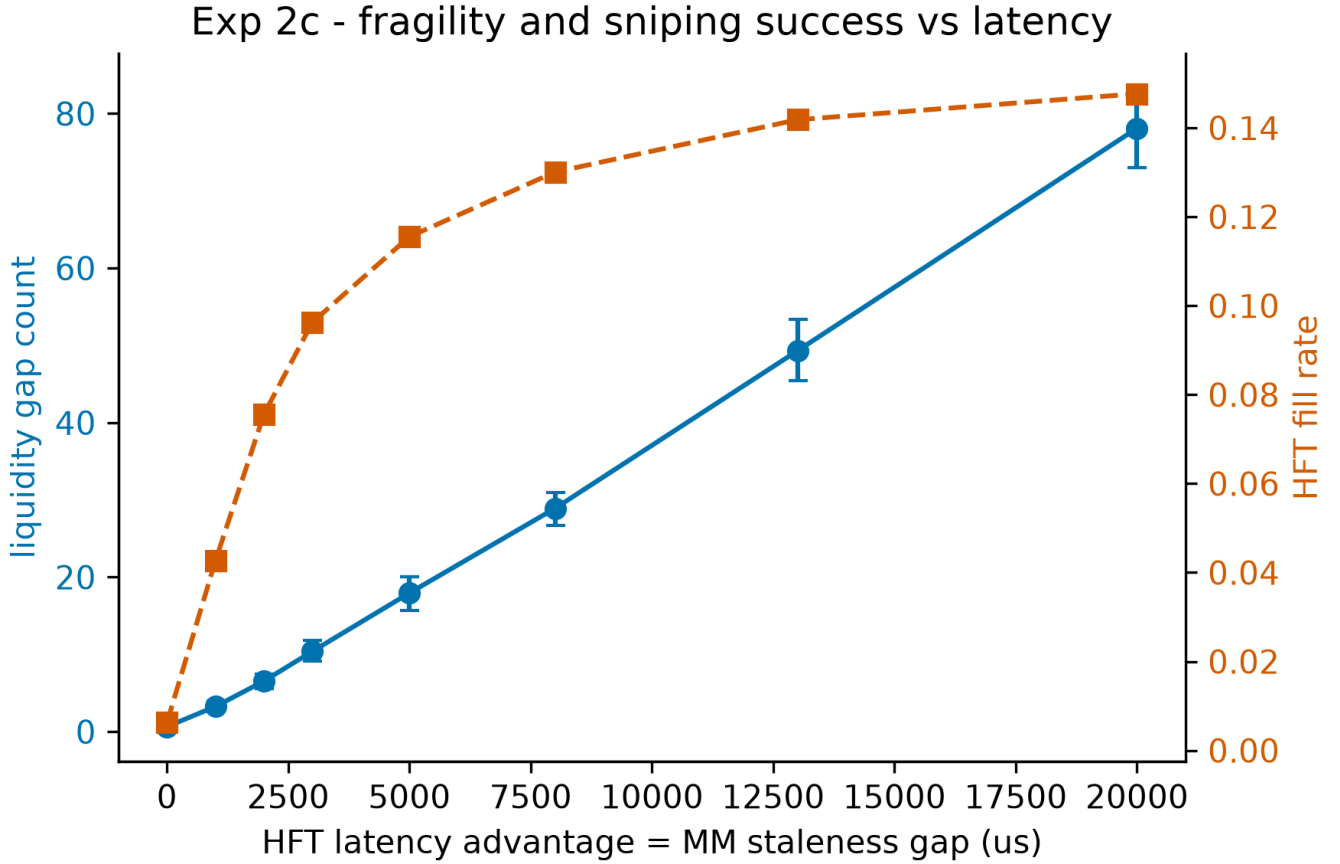


Figure 2c: Liquidity gaps and HFT fill rate versus MM staleness

Both microstructure quality measures degrade smoothly and without inflection. Liquidity gaps widen from 0.65 [0.30, 1.00] at 0 ticks to 78.10 [73.00, 83.60] at 20 ticks, while the HFT fill rate climbs from 0.00624 [0.00473, 0.00775] to 0.14765 [0.14582, 0.14945], saturating gently at the upper end of the grid. Neither series exhibits the abrupt acceleration that a phase transition would predict. Taken together, the rent, welfare, and liquidity evidence all point to the same conclusion: the latency advantage scales continuously and approximately linearly in MM staleness, and the phase-transition hypothesis is not supported by these data.

4.3 Experiment 3: HFT competition and the socially wasteful arms race

We next ask how the number of competing high-frequency traders, k , shapes the rents they extract and the costs they impose. We sweep k over the grid $\{0, 1, 2, 3, 5, 8, 13, 21\}$ with $n = 100$ seeds per cell, holding the market-maker (MM) population and order-flow process fixed. The central finding is a sharp separation between the *private* and *social* consequences of entry: individual HFT profitability collapses as competitors enter, yet the aggregate rent transferred away from liquidity providers does not. The marginal social cost of an additional HFT is therefore not a larger wealth transfer but a more fragile book.

Per-HFT rent falls monotonically and steeply, from \$34.82 [26.40, 44.84] at $k = 1$ to \$1.60 [1.21, 2.07] at $k = 21$ (Figure 3a). The decline is almost exactly hyperbolic: a one-parameter fit of the form $\text{rent}/\text{HFT} = C/k$ yields $C = 34.80$ with $R^2 = 0.99994$. (Both the C/k form and the constant are estimated from the cell means, not stored simulation outputs.) The same $1/k$ structure appears in execution shares: the per-HFT fill rate falls from 0.2886 [0.2863, 0.2909] at $k = 1$ to 0.0139 [0.0137, 0.0140] at $k = 21$, and the product $\text{fill rate} \times k$ is

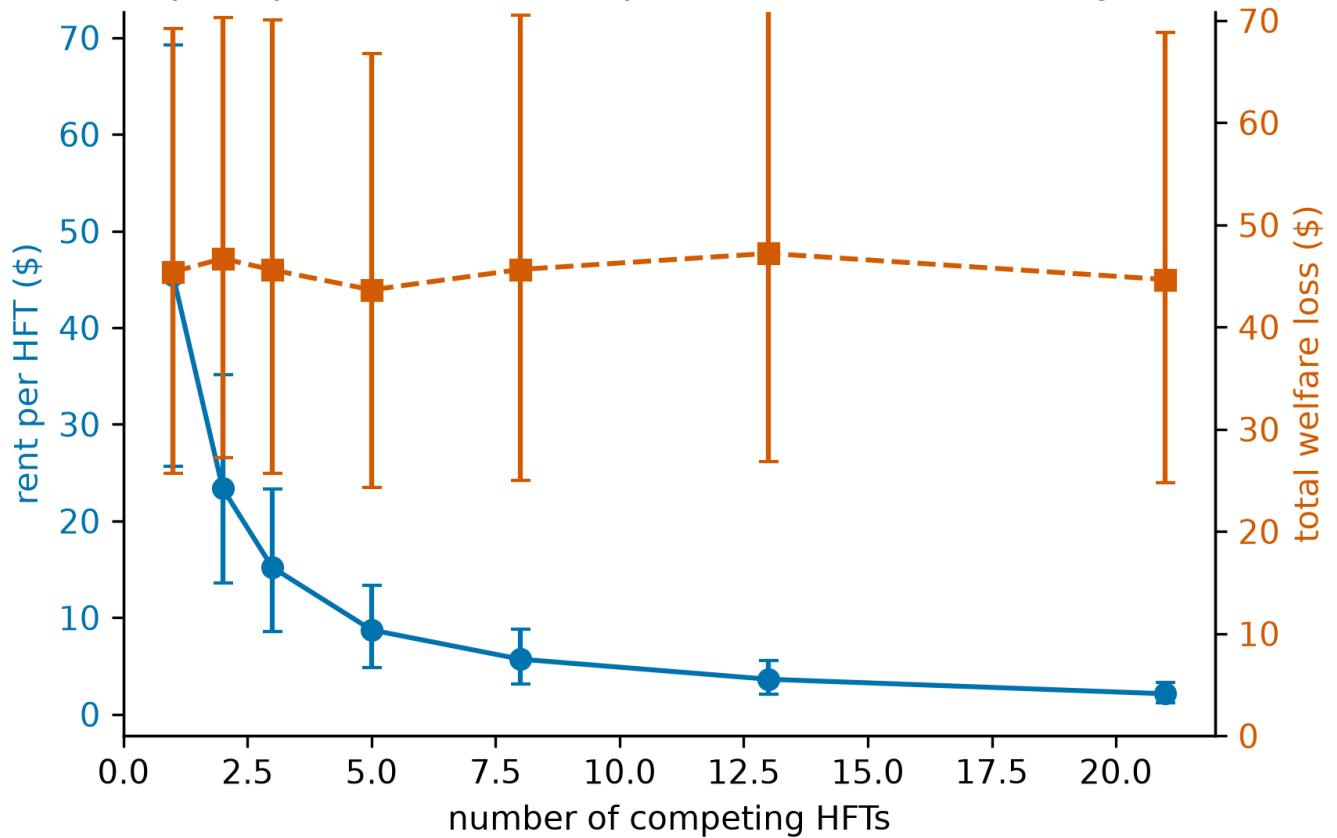
statistically flat at roughly 0.290 (derived mean 0.2896, std 0.0008 over $k = 1 \dots 21$; Figure 3c). Competition thus partitions a fixed pool of exploitable flow into ever-thinner per-agent slices without enlarging the pool itself.

The aggregate confirms this directly. Total extracted rent is flat across the entire grid (Figure 3, Figure 3b): \$34.82 [26.40, 44.84] at $k = 1$, \$34.93 [26.55, 44.87] at $k = 2$, \$34.70 [26.25, 44.80] at $k = 3$, \$33.83 [25.36, 44.03] at $k = 5$, \$34.50 [25.84, 44.76] at $k = 8$, \$34.43 [25.63, 44.87] at $k = 13$, and \$33.70 [25.39, 43.45] at $k = 21$ — a derived mean of \$34.42 (std 0.48, range 33.70–34.93). The $n = 100$ estimates make this flatness materially sharper than our earlier $n = 20$ run, which spread over 43.64–47.21 (std 1.11); the spread across the grid is now roughly a third as wide, so the constancy of total rent is a stronger claim than before, not merely a rescaled one. The transfer is cleanly zero-sum: MM PnL is approximately constant near \$73 for all $k \geq 1$ and the zero-sum residual is numerically zero at every count. We therefore decline to characterize HFT entry as raising total welfare loss; the welfare delta tracks total rent and is flat, not rising.

k	Per-HFT rent (\$)	Total rent (\$)	Fill rate	Liquidity gaps
0	—	—	—	2.84 [2.47, 3.26]
1	34.82 [26.40, 44.84]	34.82 [26.40, 44.84]	0.2886 [0.2863, 0.2909]	5.27 [4.91, 5.65]
2	17.46 [13.28, 22.43]	34.93 [26.55, 44.87]	0.1448 [0.1437, 0.1460]	7.79 [7.31, 8.29]
3	11.57 [8.75, 14.93]	34.70 [26.25, 44.80]	0.0964 [0.0957, 0.0971]	9.70 [9.15, 10.27]
5	6.77 [5.07, 8.81]	33.83 [25.36, 44.03]	0.0578 [0.0574, 0.0583]	14.92 [14.27, 15.61]
8	4.31 [3.23, 5.60]	34.50 [25.84, 44.76]	0.0362 [0.0359, 0.0365]	21.09 [20.20, 21.99]
13	2.65 [1.97, 3.45]	34.43 [25.63, 44.87]	0.0223 [0.0222, 0.0225]	32.79 [31.73, 33.91]
21	1.60 [1.21, 2.07]	33.70 [25.39, 43.45]	0.0139 [0.0137, 0.0140]	47.47 [46.22, 48.85]

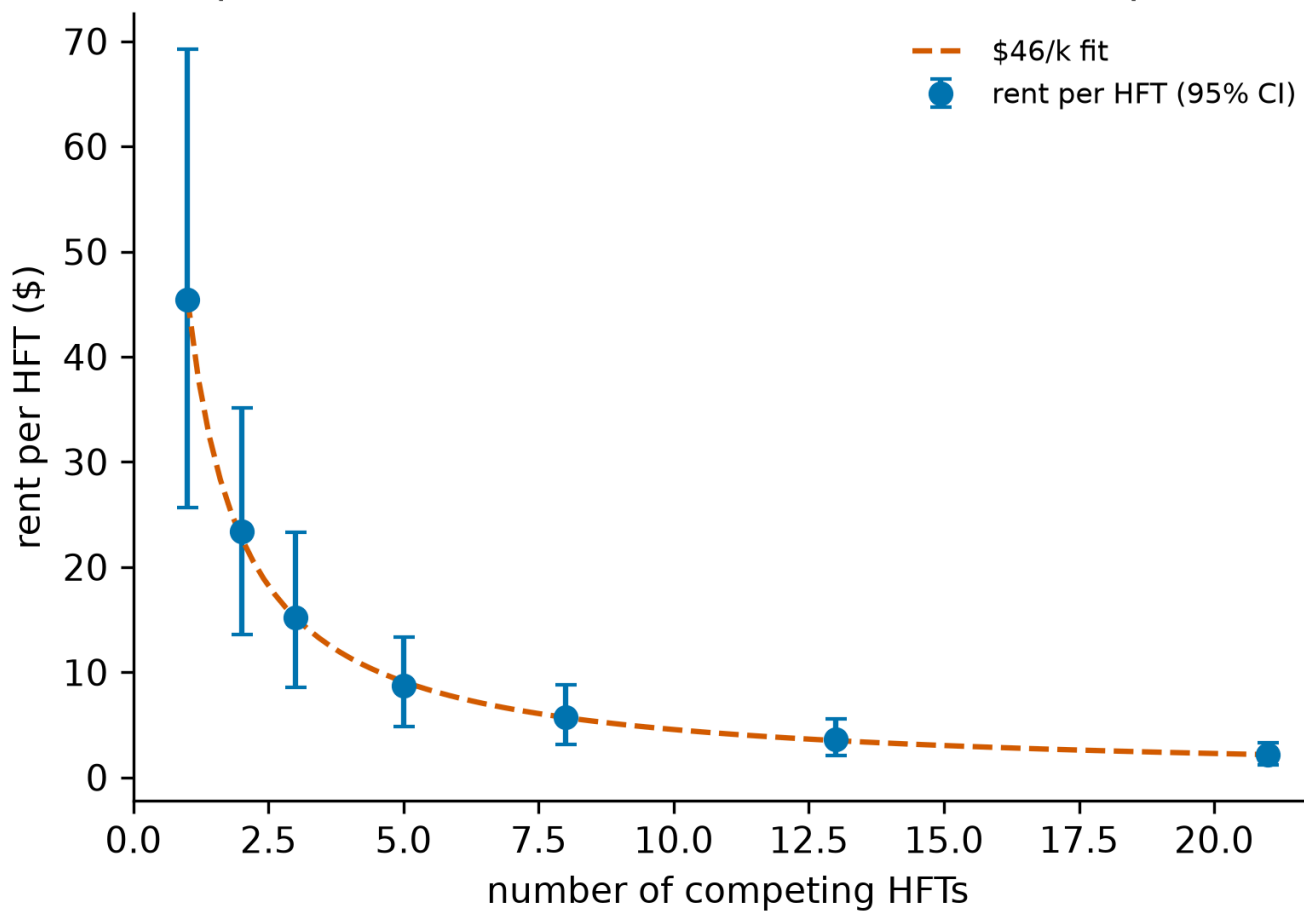
Where the marginal cost of entry does appear is in market fragility. The incidence of liquidity gaps rises steadily with k , from a baseline of 2.84 [2.47, 3.26] at $k = 0$ to 47.47 [46.22, 48.85] at $k = 21$ — roughly a 16.7-fold increase (Figure 3b). The growth is well described as linear in k (derived fit: slope = 2.13, intercept = 3.74, $R^2 = 0.99730$). This is the substantive welfare result of the experiment: each additional HFT competes away its predecessors' per-agent rent without enlarging the total transfer, but contributes an approximately constant increment to the rate at which the book is left empty. The arms race is socially wasteful not because it grows the rent pie but because the competition to capture a fixed pie progressively thins the liquidity available to ordinary participants.

Exp 3 - per-HFT rent collapses $\sim 1/k$; total loss stays flat



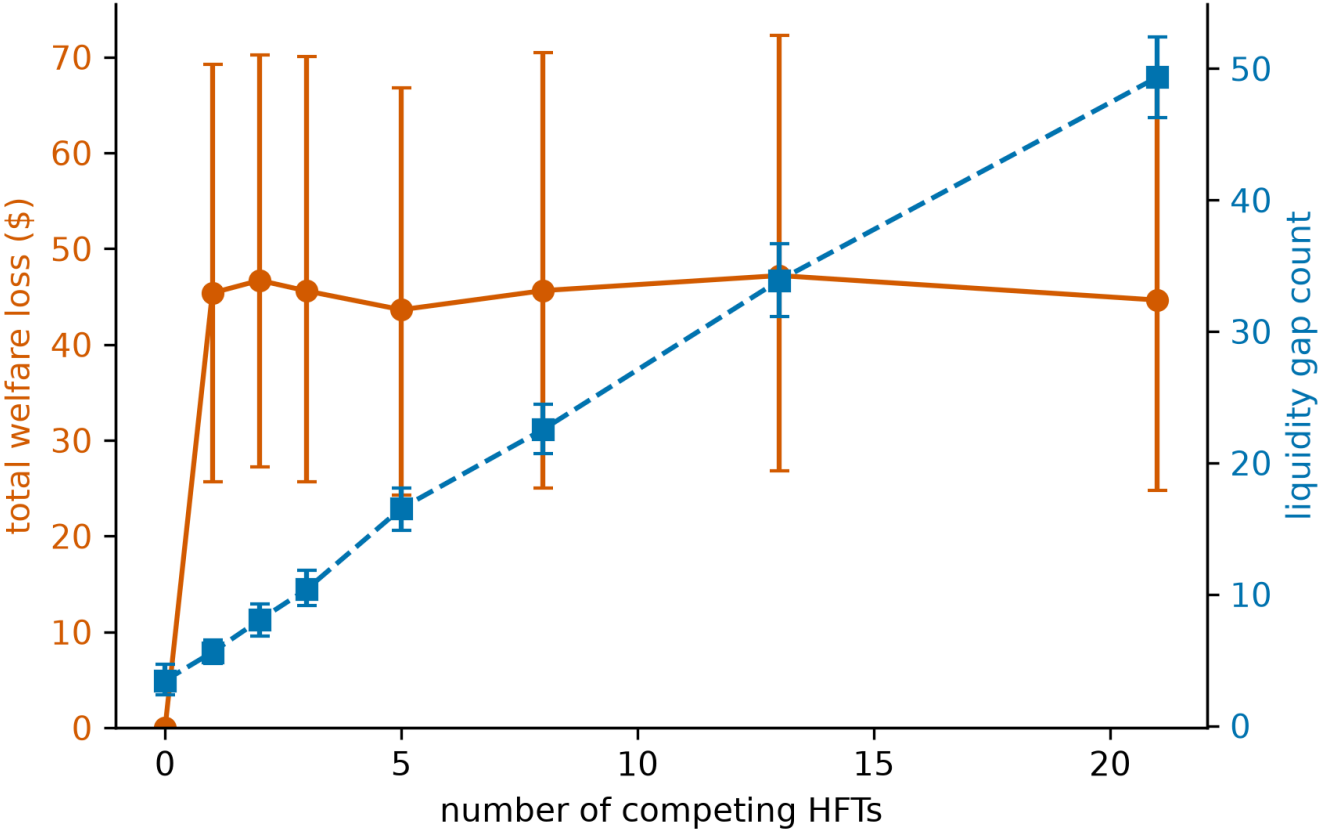
Per-HFT rent decline versus flat total welfare loss

Exp 3a - individual HFT rent erodes with competition



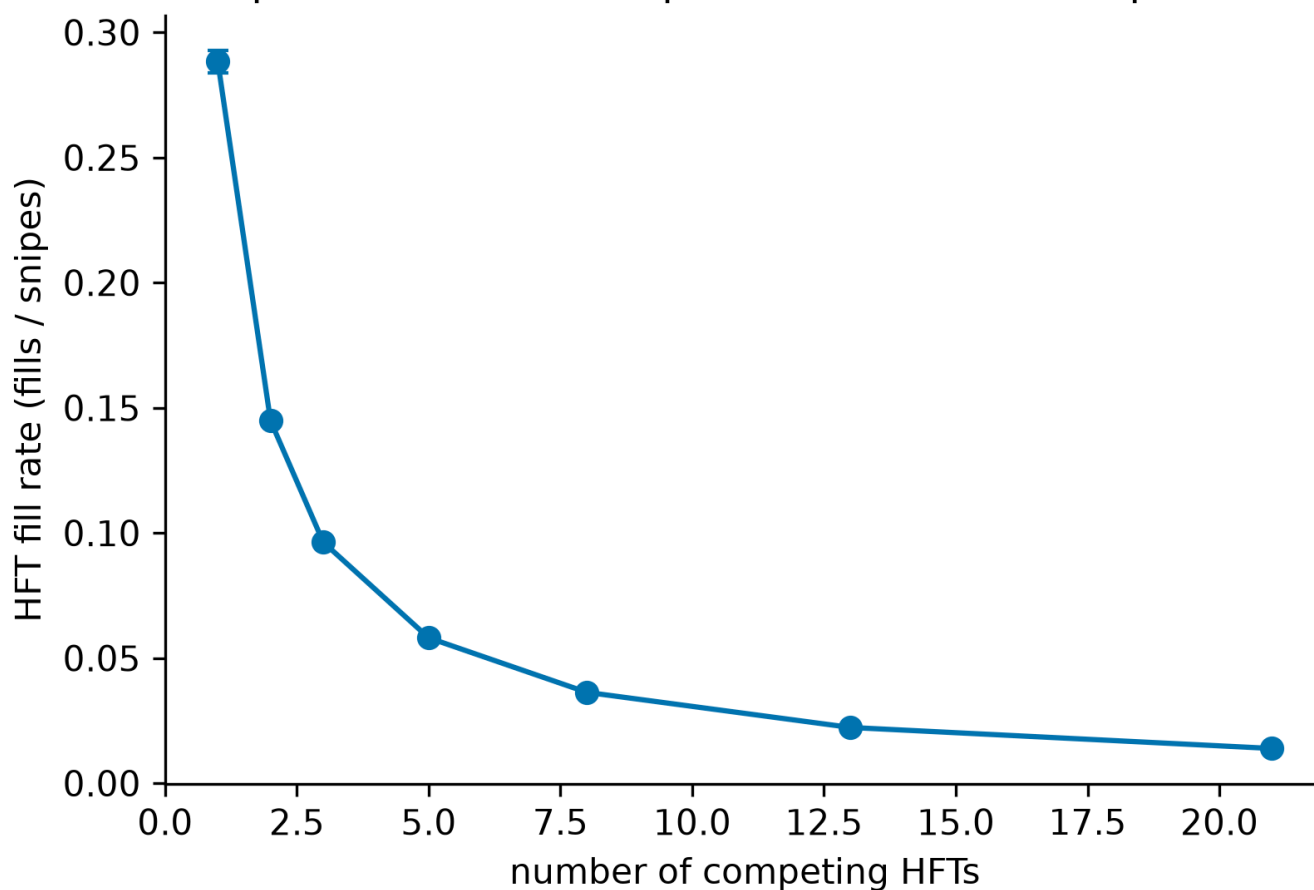
Per-HFT rent with C/k fit

Exp 3b - flat rent, but fragility rises with HFT count



Flat total rent alongside rising liquidity gaps

Exp 3c - race success per HFT falls with competition



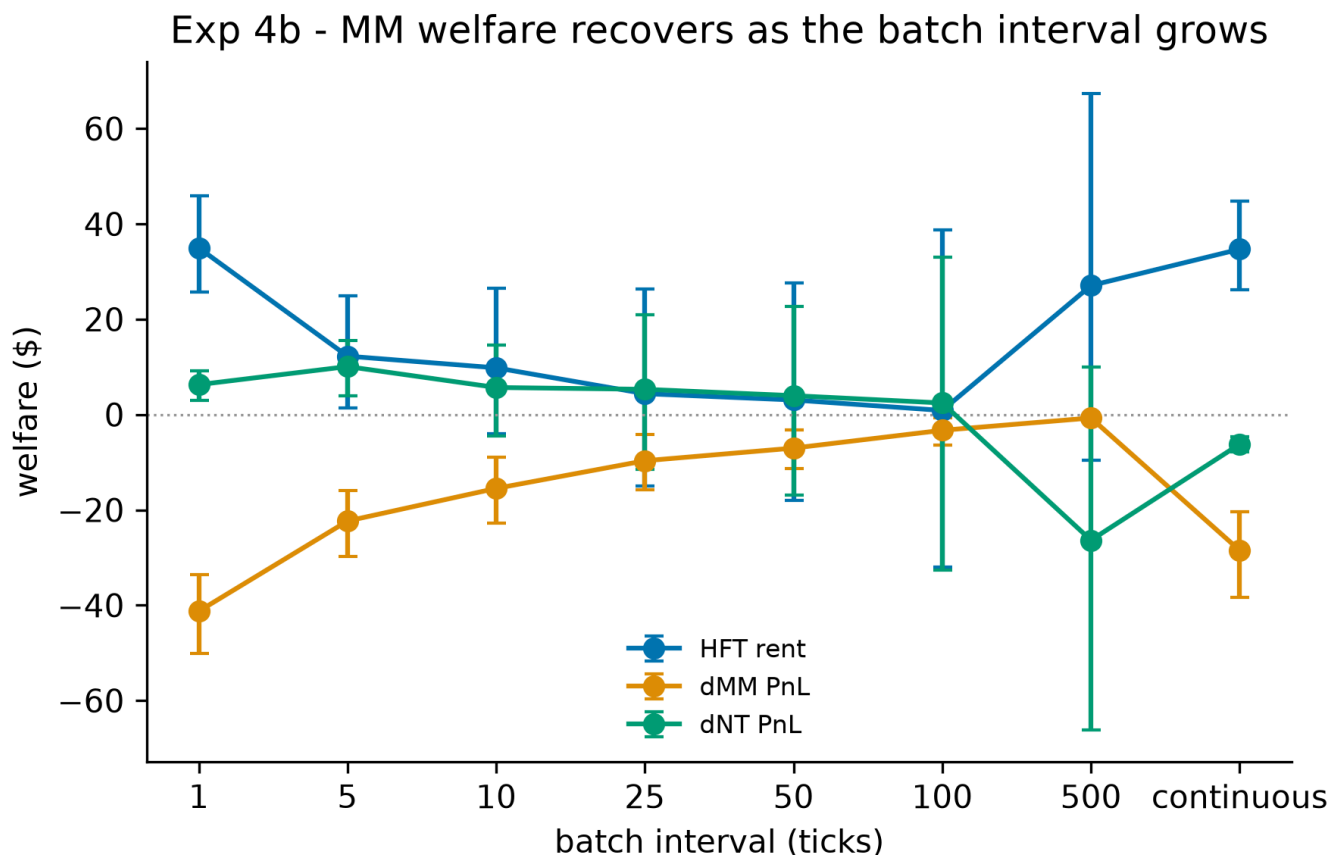
Per-HFT fill rate declining as $1/k$

4.4 Experiment 4: Batch auctions as a remedy

Having documented in Experiments 1–3 that continuous-time matching concentrates losses on the slow market maker (MM loss of 28.5) while conferring latency rents on the fast trader (HFT rent of 34.7) and leaving liquidity gaps (9.7), we now ask whether frequent batch auctions—uniform-price call auctions run at a fixed clearing interval—repair the damage. We sweep the clearing interval across $\{1, 5, 10, 25, 50, 100, 500\}$ ticks, with $n = 100$ seeds per arm, and decompose the outcome into MM welfare, HFT rent, and liquidity gaps. Both arms clear through the verified C++ matching engine: the continuous arm matches on arrival, the batch arms park the book in an auction state and clear through the engine’s own `uncross` (§3.2). An earlier revision re-implemented the clearing rule in Python; when checked against the engine it differed on tie-breaks across volume-maximizing prices (§3.2), and every batch-arm number below comes from the engine-cleared implementation. The continuous arm is byte-identical across the two revisions, so all movement relative to previously reported batch figures is attributable to the port.

The robust result is the recovery of market-maker welfare, shown in Figure 4b. Away from the degenerate single-tick auction, MM loss falls monotonically as the clearing interval lengthens: from 28.5 in the continuous market to 22.3 at interval 5, 15.5 at interval 10, 9.7 at interval 25, 7.0 at interval 50, 3.3 at interval 100, and 0.7 at interval 500—roughly 97% of the continuous-market loss eliminated. In recovery terms this is 21.6% at interval 5, 45.6% at interval 10, 66.0% at interval 25, 75.3% at interval 50, 88.4% at interval 100, and 97.5%

at interval 500. The threshold for at-least-half recovery is interval 25; the threshold for at-least-90% recovery is interval 500. The monotonicity and the size of the effect make this the headline finding, and it is the one result in this experiment that both the move to $n = 100$ and the engine port of the batch arms (§3.2) leave qualitatively untouched.



Welfare decomposition versus clearing interval

The effect on HFT rent points in the same direction but remains far noisier, and we continue to report it as suggestive rather than established. Figure 4a plots batch-arm rent against interval. The estimates are 12.26 [1.41, 24.89] at interval 5, 9.81 [-4.00, 26.44] at interval 10, 4.39 [-14.97, 26.39] at interval 25, 3.06 [-18.08, 27.67] at interval 50, 0.86 [-32.04, 38.82] at interval 100, and 27.07 [-9.66, 67.34] at interval 500, against a continuous-market rent of 34.70 (the single-tick arm registers 34.98 [25.72, 45.95], indistinguishable from continuous). At $n = 100$ the interval-5 arm is the one batch cell whose rent interval clears zero; every other batch interval still brackets it.

At $n = 20$ the rent-elimination fractions were non-monotone in the interval and appeared to peak at 500 (70.5%). At $n = 100$ they instead rise steadily over the range 5–100: 64.7% at interval 5, 71.7% at 10, 87.4% at 25, 91.2% at 50, and 97.5% at 100, before collapsing to 22.0% at interval 500. We initially read this as a clean dose-response relationship with a single anomalous cell at 500, and attributed the anomaly to the sparsity of clearing events — a 3,000-tick run at interval 500 clears only six auctions, so each seed's HFT P&L is the sum of six lumpy draws. That diagnosis predicted a specific remedy: longer runs, not more seeds. We tested it, and it was wrong in an instructive way.

Duration robustness. We re-ran the full Experiment 4 grid at 50,000 ticks with $n = 100$ (`exp4_batch_long.json`), which delivers 100 clearing events at interval 500 instead of six; like the 3,000-tick grid, this grid clears through the verified engine (§3.2). The predicted remedy failed outright. Rent intervals did not tighten — they widened to the point of collapse: at 50,000 ticks **no batch interval's rent CI excludes zero** (interval 10 spans $[-1024, 556]$; interval 500 spans $[-2250, 1894]$), and the point estimates at intervals 5–25 come out negative ($-170, -252, -194$) against a continuous-arm rent of 578 [449, 709]. The short-run elimination ordering does not reproduce in any form. The continuous arm, by contrast, behaves exactly as accumulation predicts — 578 is $16.67 \times$ the 3,000-tick rent of 34.70, with a proportionally tight interval — so the pathology is specific to rent under sparse uniform clearing, not to long runs as such. (An earlier, pre-port version of this grid, produced by the retired Python-cleared arm, had appeared to show partial tightening — intervals 10 and 25 joining 5 in excluding zero — alongside a reversed elimination ordering; neither survives the engine port, which we take as further evidence that these cells measure noise.)

The diagnostic that resolves this is the scaling of the per-seed rent dispersion. If per-run rent were an accumulation of many independent per-auction increments, its standard deviation should grow as the square root of run length (a factor of 4.08 for our 16.67-fold duration increase). It does not: across the grid the observed factors are 12.2, 48.9, 51.7, 34.6, 25.3, 22.5, and 52.9, i.e. at least linear in duration everywhere and two to three times worse than linear at intervals 5–25 and 500. A duration-proportional variance component of this kind is not auction-granularity noise. The distributional shape sharpens the puzzle: at intervals 100 and 500 the median per-seed rent is strongly negative (-495 and -2528) while the mean is small relative to the dispersion (426 and -260), i.e. a typical run loses money and the mean is carried by a thin right tail of large winners. Notably, these medians are almost unchanged from the pre-port grid (-502 and -2517): the typical-run loss structure is robust to which matcher clears the batch, and it is the extreme tails — where the mean lives — that the tie-break difference reshuffles.

The natural candidate explanation is the HFT's terminal inventory being marked against a fundamental random walk whose displacement also grows with the run. We tested it and can rule it out as the dominant term. We instrumented the agents with a weighted-average cost basis (`metrics/position.py`) that splits P&L into a realised round-trip component and an open-position mark-to-market component, and re-measured. Removing the inventory term reduces the per-seed dispersion only modestly — at interval 500 the standard deviation falls from 10553 to 8788 over 50,000 ticks, about 17% — and leaves the signal-to-noise ratio at that interval an order of magnitude below usability. The realised component itself exhibits the same super-linear growth in run length (a factor of 58 against the 4.08 that independent per-auction accumulation would predict), so the excess variance is intrinsic to round-trip P&L under sparse uniform clearing rather than to inventory carry. We report the decomposition because it is diagnostic, and because `hft_rent_realized` is a modestly cleaner measure where signal-to-noise is already adequate (it improves the continuous arm from 0.86 to 0.91), but it does not rescue the long-interval cells.

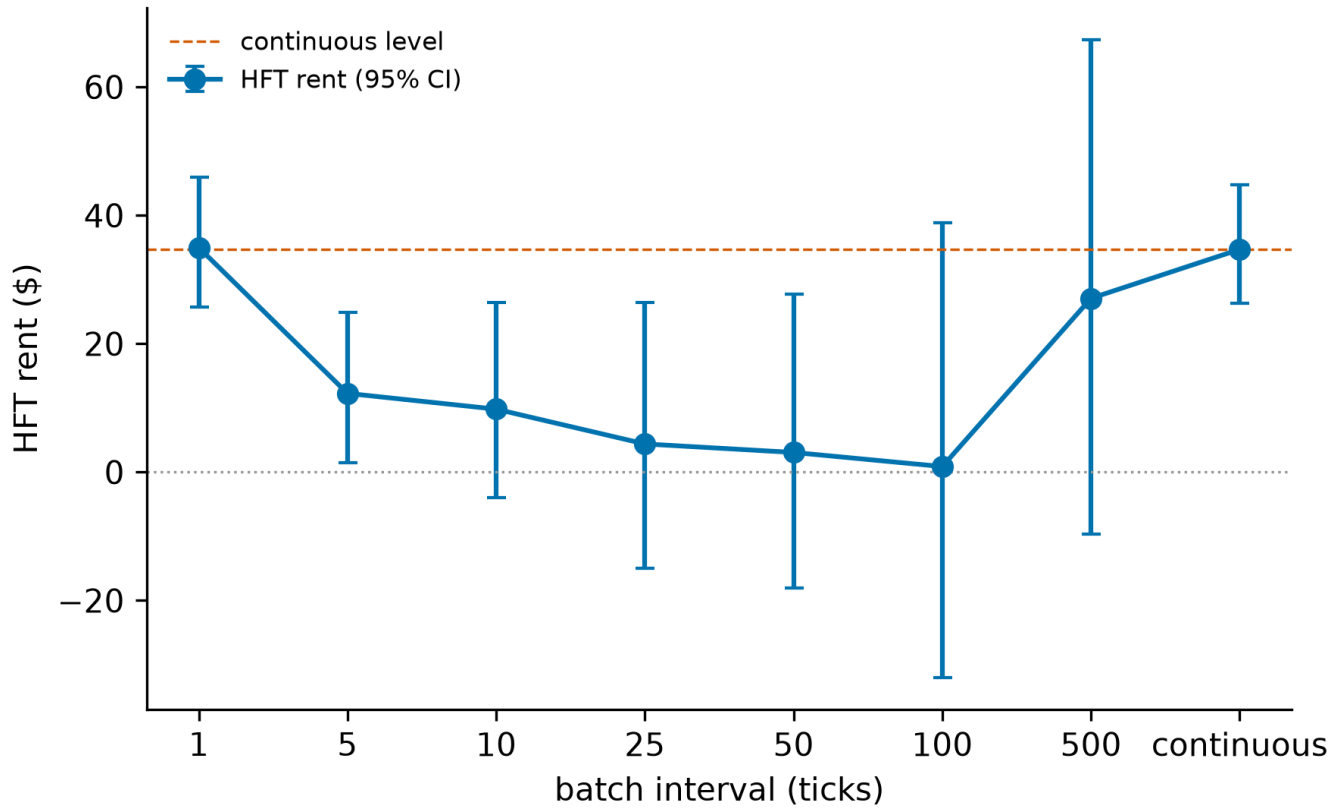
The variance source is instead the fundamental's own volatility. Holding the interval at 500 and the run at 50,000 ticks while shrinking σ from 20 to 2.5 collapses the per-seed rent dispersion from 10553 to 964 (`exp4_sigma_sweep.json`), scaling approximately as $\sigma^{1.1-1.2}$. (Since $\sigma = 20$ is the calibrated volatility of §3.7, this sweep runs from the realistic end downward: its quieter cells buy precision by leaving the regime the data supports.) (A pre-port version of this sweep had suggested $\sigma^{3/2}$; the engine-cleared re-run — whose $\sigma = 20$ cell reproduces the long grid's interval-500 rows exactly — softens the exponent while confirming the mechanism.)

But identifying the mechanism supplies no remedy, because the rent signal does not emerge as the noise recedes: the point estimate is negative at every volatility we tested (-260 , -828 , -374 , -283 at $\sigma = 20, 10, 5, 2.5$), and the signal-to-noise ratio rises only from 0.03 at the operating point to 0.29 at $\sigma = 2.5$ — where the marginally detectable quantity is a small rent *loss* (-283 against a seed-level standard error of 96), not a rent. Long-interval batch rent is not a quantity our design measures poorly; at every volatility it is either unmeasurably small or measurably absent, and no reweighting of the estimator changes that.

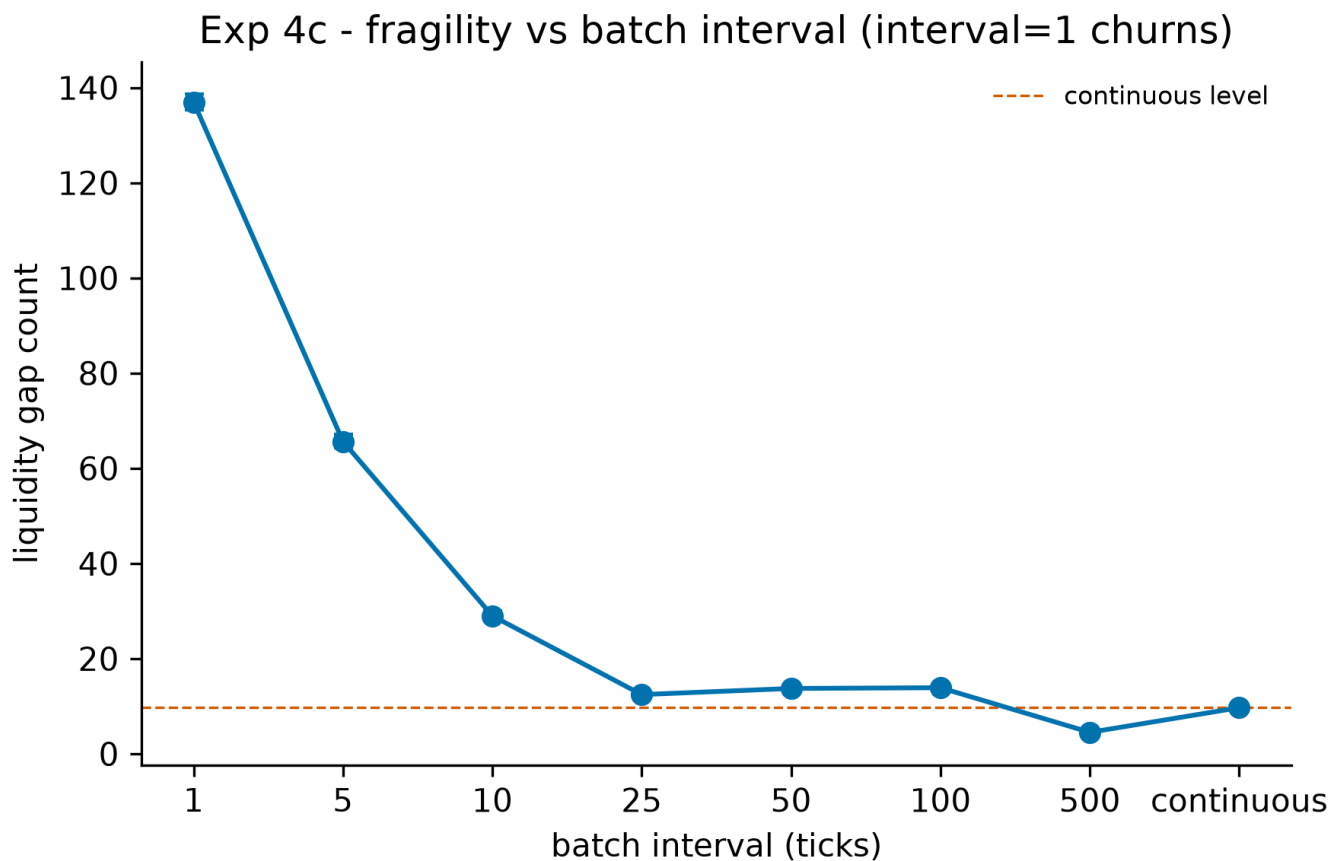
The implication is that our `hft_rent` metric does not measure the same thing at $3,000$ and $50,000$ ticks, and having tested the two obvious correctives we do not have a third to offer. We therefore treat the long-interval rent estimates as uninformative rather than merely imprecise. Closing this gap would require variance reduction of a kind our design does not currently admit — the matched-seed pairing that sharpens the MM-welfare delta cannot help here, because the no-HFT baseline has zero rent by construction and so offers nothing to difference against. We flag this as the primary open measurement problem in Experiment 4, and note that it bounds what any simulation of this form can say about rent at sparse clearing intervals, not merely what ours says.

Accordingly, we report the batch-arm rent results only where the estimator is credible — the $3,000$ -tick grid, where interval 5 is the single batch cell whose rent interval excludes zero — and we draw no rent conclusion at all from the $50,000$ -tick grid, where every batch cell brackets zero. We specifically decline to nominate an optimal interval on rent grounds. The MM-recovery result, built on the matched-seed delta, is the robust one, and the two durations tell one consistent story. At $3,000$ ticks recovery is monotone in the interval ($21.6\% \rightarrow 97.5\%$). At $50,000$ ticks it saturates: every interval ≥ 5 recovers effectively all of the market maker's loss (point recoveries 99 – 165% of the continuous-arm loss of 687), and at intervals 10 and 25 the matched-seed MM delta is significantly *positive* ($[81, 803]$ and $[55, 576]$) — the maker does measurably better with HFTs present than without at those cadences, plausibly because neutralized snipers still submit order flow the maker earns the half-spread on. We report that overshoot as an observation rather than build on it, given how dispersed the long-run levels are. The single-tick auction backfires significantly at both durations (-44.8% and -14.0% respectively).

Exp 4a - batching reduces HFT rent (point estimate)



HFT rent versus clearing interval



Liquidity gaps versus clearing interval

A per-tick call auction (interval 1) does not merely fail to help—it backfires. MM loss rises to 41.3, worse than the continuous market’s 28.5, and liquidity gaps spike to 136.97 against the continuous baseline of 9.70 (Figure 4c). The backfire remains larger at $n = 100$ than the $n = 20$ runs suggested: the MM is 44.8% worse off than under continuous matching, against the earlier $n = 20$ estimate of 35.8%. The mechanism is intuitive: a slow market maker that cannot refresh quotes within a single-tick clearing cadence is churned by the auction, repriced against before it can react, so the remedy at its most aggressive setting reproduces and amplifies the disease. Once the interval lengthens enough for the MM to participate meaningfully, liquidity gaps fall sharply—65.65 at interval 5, 28.99 at interval 10, 12.45 at interval 25—and then stabilize at a low level (13.74, 13.90, and 4.52 at intervals 50, 100, and 500). The policy reading is correspondingly conditional: batch auctions restore market-maker welfare substantially and improve depth, but only above a minimum cadence; set the interval too short and the intervention is counterproductive.

The following table collects the arm-level estimates.

Clearing interval (ticks)	MM loss	MM welfare recovery	HFT rent (point [lo, hi])	Rent elimination	Liquidity gaps	Auctions per run
Continuous	28.50	—	34.70 [26.25, 44.80]	—	9.70	—
1 (degenerate)	41.26	-44.8% (backfires)	34.98 [25.72, 45.95]	-0.8%	136.97	3000
5	22.34	21.6%	12.26 [1.41, 24.89]	64.7%	65.65	600
10	15.49	45.6%	9.81 [-4.00, 26.44]	71.7%	28.99	300
25	9.69	66.0%	4.39 [-14.97, 26.39]	87.4%	12.45	120
50	7.03	75.3%	3.06 [-18.08, 27.67]	91.2%	13.74	60
100	3.30	88.4%	0.86 [-32.04, 38.82]	97.5%	13.90	30
500	0.71	97.5%	27.07 [-9.66, 67.34]	22.0%	4.52	6

Notes: $n = 100$ seeds per arm. Both arms are generated by the verified C++ engine; the batch arms park the book in an auction state and clear through `OrderBook::uncross()` (§3.2). Brackets give 95% bootstrap confidence intervals (asymmetric); of the batch arms only interval 5 has a rent interval excluding zero. The minimum interval for at-least-50% MM recovery is 25 and for at-least-90% is 500; the minimum interval for at-least-50% rent elimination is 5. The final column reports the number of clearing events in a 3,000-tick run. A companion 50,000-tick grid at the same seed count, also engine-cleared, is reported in the duration-robustness discussion above; it shows the MM recovering fully at every interval ≥ 5 while every batch cell's rent interval brackets zero, and we there draw no long-duration rent conclusion at all.

4.5 The Hawkes endogeneity diagnostic: pilot, redesign, and a placebo-validated null

Experiment 5 asks whether self-excitation in order flow, summarized by the Hawkes branching ratio $n(t)$, rises ahead of liquidity gaps relative to normal trading. The intuition is that elevated endogeneity — orders begetting orders — could precede the quote thinning documented in earlier experiments. We report the preliminary pilot first, its design tensions second, and then the redesigned experiment that resolves them; the arc from suggestive pilot to placebo-validated null is itself part of the finding.

We fit the branching ratio over rolling windows of 500 trades (approximately 716 ticks) at a stride of 100 trades, across a 5-simulation pilot (each simulation 50,000 ticks, roughly 35,000 trades, yielding on the order of 333–338 windows per simulation). Windows whose 500-tick lead interval terminates immediately before a liquidity gap are labelled pre-gap; all others are labelled normal. Gaps are frequent in these runs — roughly one per 300 ticks, or about 173 gaps per 50,000-tick simulation, with 865 gap events in total. This labelling

yields an imbalanced split of 1,371 pre-gap windows against 303 normal windows. Because the two groups are independent rather than matched, we use the Mann-Whitney U test, correcting the paired Wilcoxon test specified in the pre-registered plan; pairing is not well defined once windows are pooled across overlapping gap neighbourhoods.

The pre-gap branching ratio is directionally consistent with the hypothesis but the difference is not statistically significant.

Quantity	Estimate
Pre-gap branching ratio $n(t)$, mean	0.0039
Normal-period branching ratio $n(t)$, mean	0.0030
Mann-Whitney U statistic	216647.0
Mann-Whitney one-sided p-value	0.12
Rank-biserial correlation (effect size)	0.043 (negligible)
Pre-gap windows (n_{pre})	1,371
Normal windows (n_{normal})	303
Pilot simulations (n_{sims})	5

Mean $n(t)$ is higher before gaps (0.0039) than in normal periods (0.0030), the expected sign, but the one-sided Mann-Whitney test does not reject the null ($p=0.12$) and the rank-biserial effect size of 0.043 is negligible. Bootstrap confidence intervals are deferred to the full run; with only five pilot simulations, interval estimates would be uninformative, so we report point estimates here and adopt the “point [lo, hi]” convention from the full study onward.

Two design tensions temper any reading of these numbers. First, the calibration window (500 trades, about 716 ticks) is coarser than the gap spacing (about one gap per 300 ticks). A window that should characterize a single pre-gap regime therefore straddles multiple gaps, blurring the contrast and producing the lopsided 1,371-versus-303 labelling. Second, the overall branching ratio is small ($n(t)$ about 0.003), implying trade arrivals here are close to Poisson. A near-zero branching ratio leaves little self-excitation for the test to detect and suggests that trade-arrival Hawkes intensity may be the wrong endogeneity proxy for what is, mechanically, a quote-thinning phenomenon; a process defined on quote updates or cancellations is the more natural candidate.

The redesigned experiment (`analysis/exp5_redesigned.py`, `exp5_redesigned.json`) implements both corrections plus a falsification control, at 10 simulations per arm $\times$ 50,000 ticks. Windows are time-based (200 ticks wide, stride 50, lead 100 — sized below the $\sim$ 300-tick gap spacing), and windows that *contain* a gap start are dropped rather than left to pollute the normal pool, which repairs the pilot’s lopsided labelling: the split becomes 1,416 pre-gap against 3,614 normal. Two event processes are fitted in every window — trade arrivals (the pilot’s proxy) and per-side best-quote updates (the natural proxy for a quote-thinning phenomenon). The placebo arm repeats the entire pipeline on fresh simulations with large exogenous

fundamental jumps ($\pm 50\sigma$) injected at known times, and points the same labelling machinery at the shock times; shocks are unpredictable from inside the market, so any “pre-shock elevation” would expose the labelling as artifactual.

Arm / process	pre mean $n(t)$	normal mean $n(t)$	p (one-sided)	rank-biserial	$n_{\text{pre}} / n_{\text{normal}}$
Treatment / trades	0.0162	0.0167	0.258	0.012	1,416 / 3,614
Treatment / quotes	≈ 0	≈ 0	0.937	-0.029	1,260 / 3,206
Placebo / trades	0.0111	0.0168	0.895	-0.038	373 / 8,614
Placebo / quotes	≈ 0	≈ 0	0.491	0.001	335 / 7,707

The placebo is null on both processes, validating the pipeline. And with sound labelling, the treatment is null too: the pilot’s directional hint does not survive its own corrections. Pre-gap and normal branching ratios are indistinguishable on trade arrivals ($p=0.258$, effect size 0.012), and the quote-update process shows essentially zero self-excitation anywhere — mechanically sensible, since the market maker’s requotes track the exogenous fundamental random walk. We conclude that at this operating point the liquidity gaps of Experiments 1–3 are *not* preceded by rising order-flow self-excitation: they arise from the adverse-selection spread-widening mechanism directly, not from a reflexive cascade, and this market’s arrival processes are close to Poisson by construction (branching ≈ 0.017 on trades, some $50\times$ below criticality), leaving little endogeneity to detect. This is a negative result about the simulated mechanism, stated as such — it is not evidence against reflexivity in real markets, where self-excitation is well documented (Filimonov & Sornette 2012). What the redesign contributes is a validated instrument: a windowing, labelling, and placebo pipeline that demonstrably does not manufacture endogeneity where none exists, ready for the real-data leg — which now awaits only a depth-depletion gap definition suited to a book that never empties. No figure accompanies this result.

5. Discussion

The four completed experiments describe a consistent picture of latency competition on a continuous limit order book whose matching layer is model-checked (the Hawkes diagnostic of §4.5 adds a placebo-validated null: the gaps are not preceded by rising order-flow self-excitation at this operating point). The arms race is a transfer rather than a destruction of value (§4.1): HFT rent of \$34.70 [26.25, 44.80] is borne predominantly by the market maker (MM PnL delta \$-28.50 [-38.31, -20.35]), with a smaller but measurable noise-trader delta (\$-6.21 [-7.87, -4.61]), all under a conservation residual indistinguishable from floating-point noise (mean 2.8×10^{-12}). That transfer scales smoothly in market-maker staleness, and we find no evidence of a critical threshold (§4.2; slope \$10.460 per tick, $\Delta\text{BIC} = -0.17$ — a magnitude too small to discriminate between the linear and segmented specifications, so the finding is the absence of a detectable knee rather than positive support for linearity). As competitors multiply, per-HFT rent dissipates as roughly $1/k$ while the *total* extracted rent stays flat near \$34.4, so the marginal social cost of an additional HFT surfaces not as a larger wealth transfer but as a more fragile book (§4.3; gaps rising approximately linearly, $\approx 2.1k + 3.7$, about $17\times$

by twenty-one HFTs). Batch auctions recover market-maker welfare monotonically away from the degenerate single-tick corner (§4.4), though their effect on HFT rent remains statistically unresolved at long clearing intervals. Below we draw out what these findings do and do not license, state their limitations plainly, and set out the work needed to harden them.

5.1 Implications for market design

The most directly actionable result is the recovery of market-maker welfare under frequent batch auctions (§4.4, Figure 4b). Switching the clearing rule from continuous matching to a uniform-price call auction recovers market-maker welfare monotonically once the clearing interval is long enough for the slow quoter to participate: at-least-50% recovery by an interval of about 25 ticks, and about 97% recovery (MM loss \$28.5 → \$0.7) by an interval of about 500 — and over 50,000-tick runs every interval ≥ 5 recovers effectively all of the loss (§4.4). This is the robust policy signal in our data. The accompanying HFT-rent reduction points in the same direction but is suggestive only, and is confined to short runs and short intervals: at 3,000 ticks only the interval-5 cell's rent interval excludes zero, and at 50,000 ticks the rent estimator's volatility-driven dispersion swamps every batch cell entirely (§4.4). We therefore decline to claim that batching reliably eliminates latency rent, and decline to nominate an optimal interval. The defensible recommendation is therefore framed in terms of *who is protected* (the liquidity provider) rather than in terms of a guaranteed reduction in the sniper's take. That framing has independent theoretical support: in Baldauf and Mollner (2020) the comparison of remedies likewise turns on protecting resting liquidity — whether by batching or by asymmetric delay — rather than on extinguishing the snipers' rent directly.

The same experiment carries an equally important warning about *how* batching is implemented: a too-short interval backfires. The per-tick auction (interval 1) does not merely fail to help — it amplifies the disease, raising MM loss to \$41.3 (worse than the continuous \$28.5, a 45% deterioration) and spiking liquidity gaps to about 137 (against a continuous baseline of 9.7). A market maker that cannot refresh quotes within a single-tick clearing cadence is churned by the auction and repriced against before it can react. The policy reading is correspondingly conditional: a batch auction is a remedy only above a minimum cadence, and a regulator or venue setting the interval too aggressively reproduces and amplifies the pathology it sought to cure. Choosing the cadence is itself a design parameter that must be matched to the speed at which honest liquidity providers can revise quotes.

The HFT-count result reframes how policy should reason about the arms race (§4.3). Because per-HFT rent is competed away as $1/k$ while total rent stays flat, one might be tempted to conclude that competition is self-correcting — that entry erodes the prize until further entry is unprofitable, with no residual harm. Our data reject that reading. The cost does not vanish; it changes form, surfacing as fragility under a flat aggregate rent — liquidity gaps growing roughly linearly in the number of competitors, about $17\times$ by twenty-one HFTs. The arms race is socially wasteful precisely because competition to capture a fixed pool of exploitable flow progressively thins the depth available to ordinary participants. Policy therefore cannot rely on rents self-limiting as a substitute for structural reform: an observation that individual HFT margins are thin says nothing about the externality the competition imposes on book quality, and a regulator watching only realized rents would see a stable, even reassuring, aggregate while the market grew steadily more brittle beneath it.

Finally, a methodological point bears on how much weight these design conclusions can carry. Because the matching layer is model-checked in TLA+ (§3.1) and every welfare decomposition closes through a machine-precision zero-sum residual (§4.1), the emergent arms-race dynamics — the rent transfer, the liquidity gaps, the crossover into market-maker losses — cannot credibly be attributed to a design-level bug in order-priority handling, partial-fill accounting, or event sequencing on which the agents trade (within the verification scope stated in §1.2). This removes a long-standing confound in agent-based market models, where an emergent pathology may reflect either the economics under study or an artifact of the bespoke, unverified simulator. For a market-design audience, the payoff is that both arms of the clearing-rule comparison now run on a substrate whose correctness is certified — the batch arm clears through the engine’s own `uncross` (§3.2) — so the burden of explaining the observed effects falls on agent incentives and the clearing rule, which is exactly where the design argument needs it to fall. The tie-break divergence uncovered while closing that seam (§3.2) is the cautionary half of the same point: our own source-verified Python replica of the clearing rule differed from the verified engine the first time the two were results-checked.

5.2 Limitations

Several limitations bound the strength of these conclusions and should be read as constraints on external validity, not as incidental caveats.

First, the agents are simulated and stylized rather than estimated from market data. The environment they trade in is now disciplined by real order flow (§3.7: arrival intensity and fundamental volatility fitted to BTCUSDT-perp moments, with both fits verified by simulation), but the strategic structure — the market maker’s quoting rule, the snipers’ race, the constant order sizes — remains designed rather than estimated, and the sim’s spread and depth levels sit far from the perp’s (§3.7). The 3×3 sensitivity-by-decay robustness sweep (§4.1), in which HFT rent is significantly positive and the zero-sum identity holds in all nine cells (rent range \$16.56–\$46.63), partly mitigates the concern that the headline depends on a single hand-tuned cell, but it does not substitute for estimation against real microstructure. The magnitudes we report should accordingly be read as internally consistent quantities of a controlled experiment, not as forecasts of rents or gap counts in any live market.

Second, the retired Python matcher left a measurable mark before it was replaced, and that mark is worth recording even though nothing now carries it. Both Experiment 4 grids — 3,000 and 50,000 ticks — clear through the verified engine’s `uncross` (§3.2). Re-running the long grid through the engine was not cosmetic: the pre-port version had shown intervals 10 and 25 excluding zero and a reversed elimination ordering, and neither survives on the engine-cleared arm (§4.4) — the tie-break divergence of §3.2 was material at long horizons, not a curiosity. With the volatility sweep also re-run through the engine (§4.4, `exp4_sigma_sweep.json`), no pre-port artifact remains in the Experiment 4 record. The sweep’s re-measured exponent — approximately $\sigma^{1.1-1.2}$ against the pre-port $\sigma^{3/2}$ — is a second instance of the same lesson: constants measured on the retired matcher did not transfer even where the mechanism did.

Third, latency resolves only at tick (`dt_us`) granularity. Because the engine timestamps events at `dt_us = 1000`, sub-tick latency offsets are not realizable, and the staleness sweep of Experiment 2 is run on a tick-aligned grid by design (§4.2). This is faithful to the resolution at which the advantage materializes in this

harness, but it means the latency axis is coarse: the detectability floor we report (rent first clearing zero at two ticks) is a property of this grid, and finer latency structure below one tick is outside what the apparatus can represent.

Fourth, the periodic-auction arm carries IOC-in-batch semantics whose interaction with uniform clearing is a modelling choice rather than an institutional given. Immediate-or-cancel orders submitted into a discrete call auction do not map onto continuous IOC behaviour, and this interaction may compound the lumpy per-batch HFT P&L that uniform clearing already induces and that inflates the rent confidence intervals (§4.4). This reinforces the decision to treat the batch-arm rent reduction as suggestive rather than established.

Fifth, the study covers a single asset on a single venue with $n = 100$ seeds. There is no cross-venue routing and no multi-asset correlation structure of the kind that motivates much of the empirical arms-race literature. The 3×3 robustness grid partly addresses the operating-point choice but not the single-venue, single-asset scope. We also note that sample size is not the binding constraint everywhere: in the long-clearing-interval cells of Experiment 4 the limiting factor is the rent estimator itself, whose dispersion grows super-linearly in run length and therefore converges under neither more seeds nor longer runs (§4.4). We tested the most obvious corrective — decomposing P&L to strip the inventory mark-to-market term — and it recovers only about 17% of the dispersion, so the problem is not inventory carry. Identifying and correcting the residual variance source is the highest-priority methodological fix for this experiment, and until it is done the rent ranking across long clearing intervals should be regarded as unmeasured rather than merely noisy.

Sixth, the Hawkes null of §4.5 is internally clean — time-based windows, both trade and quote processes, and a passing placebo — but its external reach is bounded by construction: this market's arrival processes are nearly Poisson (noise traders arrive as per-tick Bernoulli draws; HFTs react to the exogenous fundamental), so the finding that gaps are not preceded by self-excitation *here* says the simulated gap mechanism is adverse-selection-driven, not that reflexivity is absent in real markets, where it is well documented. The diagnostic instrument is validated; the real-data leg that would give the endogeneity question empirical teeth has not run, and requires a depth-depletion gap definition for books that never empty (§5.3).

5.3 Future work

The limitations above map directly onto an agenda for hardening these results.

The first priority is to complete the calibration of §3.7. The environment parameters (arrival intensity, fundamental volatility) are now disciplined by a month of BTCUSDT-perp order flow, but three extensions remain: re-running the experimental grid at the fitted parameters — where the binding change is arrival intensity, $2.8\times$ the operating point's, volatility having landed within 3% of it; exercising the now-implemented lognormal size model (§3.7) in that re-run; and enriching the market maker enough that spread and depth *levels*, not just the environment, can be brought toward the data. Only then can the rent, welfare, and fragility magnitudes be read as calibrated quantities rather than as internally consistent outcomes at a data-adjacent operating point. This remains the single change that would most strengthen the external validity of Sections 4.1–4.3.

The second priority is Experiment 5's real-data leg — the only piece of the §5.3 redesign agenda still open. The simulation side is complete (§4.5): time-based windows, quote-update and trade processes, and a passing placebo, yielding a validated null at the operating point. Running the same pipeline on the BTC tape requires

one new component: a depth-depletion definition of a “gap” suited to a book that never empties (e.g., top-k depth falling below a percentile floor), after which the windowing, labelling, and placebo machinery apply unchanged. That leg is where the endogeneity hypothesis gets its empirical test, in the Filimonov–Sornette spirit, on flow where self-excitation is known to exist.

Third, the single-venue, single-asset scope should be relaxed. A multi-venue extension would let us study latency arbitrage across correlated instruments and order routing, the setting in which the mechanical-arbitrage evidence motivating BCS is most salient, and would test whether the flat-total-rent-with-rising-fragility pattern of Experiment 3 survives when snipers can compete across books rather than within one.

Fourth, the strategy spaces of both the market maker and the HFTs are deliberately narrow and should be enriched — for example, inventory-aware or quote-fade defences for the maker and more elaborate timing or order-type strategies for the snipers — to test how far the welfare decomposition and the batch-auction recovery are robust to more sophisticated play on both sides.

6. Conclusion

We have provided what is, to our knowledge, the first experimental test of the Budish–Cramton–Shim welfare decomposition conducted on a formally verified matching engine. By driving latency-differentiated snipers, an adverse-selection market maker, and noise traders through a C++ limit order book whose continuous-matching core has been model-checked in TLA+ with zero invariant violations, and which we drive entirely through a pybind11 bridge, we turn the closed-market welfare identity into an audited, machine-precision check rather than a modeling assumption: the zero-sum residual closes to floating-point zero on every run. Because the matching layer is model-checked against its specification with zero invariant violations—now covering the continuous cross action itself, not only book management, within the bounded scope stated in §1.2—the emergent arms-race dynamics, liquidity gaps, and fragility we report are attributable to rational latency competition on the continuous-clearing design rather than to design-level engine artifacts.

Within this conserved frame three headline results emerge, each stated with its honest nuance. First, HFT latency-arbitrage rent is a near-exact zero-sum transfer from liquidity providers, borne predominantly by the market maker (about 82% of the transfer) with a smaller but statistically measurable cost to noise traders. Second, the social cost of speed scales smoothly: rent rises with market-maker staleness at a near-constant marginal rate and we detect no phase transition—though we emphasize that at our sample size the model comparison is close to indifferent between linear and segmented specifications, so this is an absence of evidence for a knee rather than evidence of its absence—and as competing HFTs multiply, per-agent rent dissipates while total extracted rent stays flat, so the marginal cost of added competition surfaces as rising fragility, not rising extraction. Third, frequent batch auctions restore market-maker welfare monotonically above a minimum cadence (the per-tick auction backfires), with the parallel reduction in HFT rent only suggestive: several batch-arm rent intervals still bracket zero, and at long clearing intervals the rent estimator’s dispersion grows super-linearly in run length for reasons we have diagnosed but not yet cured, so the rent ranking across intervals is not something our measurements can settle. The methodological point subsumes them all: pairing a verified substrate with a closed welfare identity lets us claim these dynamics as economics rather than artifact, a discipline we believe the agent-based microstructure literature should adopt more widely.

Code and Reproducibility

All experiment code lives under `bcs_research/` (agents in `bcs_research/agents/`, the latency scheduler and batch-auction scheduler in `bcs_research/simulation/`, metrics in `bcs_research/metrics/`, and the experiment drivers in `bcs_research/experiments/`), with the `pybind11` bridge in `bcs_research/bindings/engine_bindings.cpp`. The matching engine is verified by the TLA+ specifications in `spec/` (notably `MatchingEngine.tla` and `Auction.tla`; see the TLC verification log in `docs/Verification.md` for the engine state-exploration record). Results are written as JSON under `bcs_research/results/experiments/` and are reproducible by re-running the experiment scripts (`exp1_hardening.py` for the Experiment 1 results, `exp1_primary.py`, the Exp 2–4 sweep drivers, and `hawkes_bridge.py` for the §4.5 supplementary Hawkes pilot) at the seeds and operating point documented in §3.6.

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